

Original Article

Elucidating the underlying therapeutic mechanism of Feikang prescription on chronic obstructive pulmonary disease: insights from a comprehensive analysis

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ABSTRACT

Background: Ranking fourth highest mortality rate globally, chronic obstructive pulmonary disease (COPD) exerts a heavy burden on both economies and healthcare systems. The Feikang prescription, a traditional Chinese herbal remedy, is utilized for its potential in managing COPD symptoms. Still, the primary components and underlying mechanisms of the Feikang prescription remain inadequately understood.

Purpose: Using Ultra-High Performance Liquid Chromatography-tandem Mass Spectrometry/MS (UHPLC-MS/MS) in conjunction with network pharmacology, molecular docking (MD), molecular dynamics simulation, surface plasmon resonance (SPR) assay and *in vivo* experimental validation, this study sought to elucidate the therapeutic effects and potential mechanism of Feikang prescription in the treatment of COPD.

Methods: UHPLC-MS/MS and network pharmacology analysis were employed to identify active components, as well as corresponding targets of the Feikang prescription and protein targets of COPD. Molecular docking was used to evaluate the interactions between the Feikang prescription's primary targets and its constituents, and molecular dynamics simulations (MDs) were performed to confirm these findings. *In vitro* SPR experiments were used to directly verify interactions. Subsequently, a cigarette smoke (CS)-induced COPD animal model was established for the investigation. Various techniques were employed, including hematoxylin-eosin (H&E) staining, Masson staining, leukocyte counts, and cell sorting in bronchoalveolar lavage fluid (BALF), enzyme-linked immunosorbent assay (ELISA), Prussian blue DAB staining, immunohistochemical staining,

Abbreviations: BALF, bronchoalveolar lavage fluid; BP, biological process; CC, cellular component; COPD, chronic obstructive pulmonary disease; CS, cigarette smoke; CSE, cigarette smoke exposure; 7-DHC, 7-dehydrocholesterol; ELISA, enzyme-linked immunosorbent assay; FEL, free energy landscape; FTH, ferritin heavy chain; FTL, ferritin light chain; GO, Gene Ontology; GPX, glutathione peroxidase; GPX4, Glutathione peroxidase 4; GSH, glutathione; HE, Hematoxylin and eosin; HP-1, Heterochromatin Protein 1; HRP, Horseradish Peroxidase; ICS, inhaled corticosteroids; IF, Immunofluorescence; IHC, Immunohistochemistry; KEGG, Kyoto Encyclopedia of Genes and Genomes; LC-MS, liquid chromatography-mass spectrometry; MD, molecular docking; MDs, molecular dynamics simulations; MF, molecular function; mIF, multiplex immunofluorescence; MLI, Mean Linear Intercept; MPO, myeloperoxidase; PDB, Protein Data Bank; PPI, protein-protein interaction; Rg, radius of gyration; RMSD, root-mean-square deviation; RMSF, root-mean-square fluctuations; ROS, reactive oxygen species; RT-qPCR, quantitative reverse transcription PCR; SASA, solvent accessible surface area; SASP, senescence-associated secretory phenotype; SLC7A11, solute carrier family 7 member 11; SPR, surface plasmon resonance; TCM, Traditional Chinese Medicine; TIC, total ion chromatogram; UHPLC-MS/MS, Ultra-High Performance Liquid Chromatography-tandem Mass Spectrometry/MS; WB, Western blotting.

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immunofluorescence (IF), multiplex immunofluorescence (mIF), quantitative real-time PCR (qRT-PCR), and Western blotting (WB) analysis *in vivo* experiments.

Results: The active ingredients and targets in the Feikang prescription were systematically identified through an integrative approach that combined network pharmacology analysis with UHPLC-MS/MS. Among the associated target proteins were Hp, IL-6, myeloperoxidase (MPO), IL-1 β , MMP9, TNF, and p53. We clarified the regulatory mechanisms involved in the modulation of glutathione peroxidase (GPX) activity, inflammation, oxidative reaction as well as the p53 and Wnt signaling pathways. The MD and MDs analysis additionally demonstrated that the Feikang prescription might elicit its therapeutic effects by stabilizing the binding of MPO with pachymic acid and p53 with apigenin, which are respectively implicated in inflammation, oxidative stress and cell senescence as well as the interaction of MPO and p53. The SPR experimental results exactly indicated the direct binding affinity among them. *In vivo* experiments confirmed the therapeutic effects of the Feikang prescription in COPD and revealed its dual capacity to inflammation-oxidative stress-ferroptosis and cell senescence. The mIF analysis validated the crosstalk between Glutathione peroxidase 4 (GPX4)-dependent ferroptosis and Wnt5a/p53-mediated cell senescence.

Conclusion: Our results address an unmet clinical need in COPD by positioning the Feikang prescription as a prospective treatment candidate, likely through mechanisms involving Wnt5a/p53-mediated cell senescence and GPX4-dependent ferroptosis.

Introduction

As a major contributor to global health and economic burden, COPD is a heterogeneous pulmonary disorder manifested as long-standing respiratory symptoms, such as cough, excessive sputum production, and breathlessness (Dransfield et al., 2023). Chronic inflammation and oxidative stress constitute fundamental pathological hallmarks of COPD, with disease pathogenesis further influenced by immunological dysregulation and protease-antiprotease imbalance (Gao et al., 2024; Wang et al., 2022; Wechsler and Wells, 2024). The three main pillars of the current therapy plan are long-acting anticholinergic drugs, long-acting β_2 agonists, and inhaled corticosteroids (ICS) (Agustí et al., 2023). Despite advances in treatment strategies, the focus remains primarily on symptom management, which fails to halt disease progression or reverse airway injury (Dransfield et al., 2023; Wang et al., 2020). This underscores the urgent need for novel treatment strategies focusing on the underlying molecular mechanisms of COPD. Delving into these mechanisms is anticipated to fill existing knowledge gaps and aid in the development of individualized treatment regimens.

Bronchial epithelial cells serve as the primary barrier against pathogens and dangerous particles in inhaled air (Domingo et al., 2025). Prolonged exposure to hazardous chemicals, especially CS, disrupts cellular growth processes and leads to cellular malfunction (GOLD, 2025). The upregulation of pro-inflammatory cytokines for IL-6, IL-8, TNF- α , and IL-1 β , alongside heightened oxidative stress, which is demonstrated by higher MPO levels, are the main causes of chronic inflammation, a hallmark of COPD (Shao et al., 2020; Zhu et al., 2024). In addition to causing airway injury, these inflammatory mediators perpetuate the oxidative stress cycle, featuring an overproduction of reactive oxygen species (ROS) and diminished antioxidant defenses (Buekers et al., 2024; Tang et al., 2021; Zhang et al., 2024b). Oxidative stress exacerbates cellular damage and triggers ferroptosis, an iron-dependent form of programmed cell death characterized by the dysregulation of key ferroptosis markers (Yan et al., 2021; Yoshida et al., 2019), such as ferritin heavy chain (FTH), ferritin light chain (FTL), GPX4, and solute carrier family 7 member 11 (SLC7A11) (Wang et al., 2025b). Recent researches have pointed to the role for ferroptosis in COPD progression, which links iron, GPX4 inactivation, ROS, lipid peroxide accumulation and oxidative stress to cellular dysfunction (Jiang et al., 2024; Yang et al., 2025; Yoshida et al., 2019).

Concurrently, several studies have found considerable cellular senescence of the bronchial epithelium in COPD (Wechsler and Wells, 2024; Zhan et al., 2025). Cell senescence, defined by a permanent halt to cell cycle arrest, has emerged as a pivotal factor in pathogenesis of COPD (Di Micco et al., 2021). Senescent cells proliferate in the lungs and exhibit elevated levels of senescence-associated markers in COPD, like senescence-associated secretory phenotype (SASP), p16, p21, p53, and

Ki-67 (Devulder, 2024; Paudel et al., 2022; Zhou et al., 2024). The Wnt signaling pathway, particularly the marker Wnt5a (Deng et al., 2022; Mastelaro de Rezende et al., 2020), has been shown to regulate cellular senescence (Chen et al., 2021; Monk et al., 2015). Emerging evidence suggests potential crosstalk between ferroptosis and cellular senescence (Sun et al., 2024; Zhou et al., 2025b), while it is yet unknown whether GPX4-dependent ferroptosis influences Wnt5a-mediated senescence in COPD. Thus, the precise mechanisms underlying this interplay remain poorly understood, particularly in context of COPD.

Traditional Chinese Medicine (TCM) formulations have demonstrated potential in alleviating COPD symptoms through their multi-component, multi-target mechanisms (Cheng et al., 2025; Pang et al., 2024; Xie et al., 2024; Xu et al., 2025), with the Feikang prescription being particularly noteworthy (Yang et al., 2020). Inflammation, oxidative stress, ferroptosis, and cellular senescence are hot topics in increasing investigation in COPD (Li et al., 2021; Wang et al., 2025a; Xie et al., 2023; Zeng et al., 2024). Nevertheless, the specific molecular pathways through which TCM exerts its therapeutic effects, particularly its role in modulating the feedback loop of inflammation-oxidative stress-ferroptosis with cellular senescence, remain unexplored.

To thoroughly examine the molecular regulatory mechanisms of Feikang prescription on airway inflammation in COPD, we employed a range of methods in this study, like UHPLC-MS/MS, network pharmacology analysis, MD, MDs, *in vitro* SPR assay integrated with *in vivo* experiments. Our investigation clarified the multi-component and multi-target properties of the Feikang prescription at various levels. We confirmed the therapeutic effect of the Feikang prescription and further described the molecular mechanisms on GPX4-dependent ferroptosis that may regulate Wnt5a/p53-mediated cell senescence, thereby reducing airway inflammation-oxidative stress-ferroptosis, and cell senescence in COPD. The trajectory of our work was displayed in **Graphical Abstract**.

Materials and methods

Feikang prescription

The Feikang prescription, provided by the Chinese Herbalist Department of Gushengtang in Guangzhou (Guangdong Province, China), consisted of 24 g of Chinese five-fingered hairy peach (*Ficus Simplicissima* Lour), 16 g of Taizi shen (*Radix et Rhizoma Ginseng*), 10 g of Bai Zhu (*Rhizoma Atractylodis Macrocephalae*), 8 g of Zi Su (*Perilla frutescens*), 8 g of Ku Xing Ren (*Semen Armeniacae Amarum*), and 12 g of Fu Ling (*Poria cocos*). After filtration and concentration to 1.4 g/ml, the crude medication decoction was stored at -20 °C.

UHPLC-MS/MS and network pharmacology

We leveraged Thermo QE Plus liquid chromatography-mass spectrometry (LC-MS) technology and UHPLC-MS/MS to profile the chemical composition of Feikang prescription samples. Subsequently the target proteins corresponding to the chemical constituents were identified through integrated queries of TCMSP database (<https://www.tcmsp-e.com/>) and Swiss TargetPrediction (<http://www.swisstargetprediction.ch/>). The UniProt database (<https://www.uniprot.org/>) facilitated gene annotation for these molecular targets. Targets associated with COPD were retrieved from the GeneCards database (<https://www.genecards.org/>) and TTD database (<https://db.idrblab.net/>).

The bioactive component targets of the Feikang prescription and the COPD-related targets were uploaded to the online platform jvenn (<http://www.bioinformatics.com.cn/static/others/jvenn/>) to create a Venn diagram. Interrogation of the STRING database (<https://cn.string-db.org/>) generated the protein-protein interaction (PPI) network from potential targets, which Cytoscape v3.9.1 transformed into topological visualizations. Functional annotation progressed through the PANTHER database (<https://pantherdb.org/>) with criteria ($\log_2|FC| \geq 1$ and $p < 0.05$), where Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) enrichment revealed potential targets of the Feikang prescription against COPD (<https://www.bioinformatics.com.cn/>).

MD verification

We executed MD through AutoDock Tools software (version 1.5.6) following a rigorous methodological workflow. First, small molecule ligands were attained from the TCMSP database and subsequently hydrogenated. The ligands were then converted into PDBQT format files. Next, we retrieved large molecule protein structures of our target genes from the UniProt database (<http://www.uniprot.org/>), while the X-ray crystal structure of the core target was downloaded from the Protein Data Bank (PDB) (<https://www.rcsb.org/>) (Bai et al., 2024; Pang et al., 2024). Solvent molecules were eliminated from the X-ray crystal structure. The proteins were pre-processed by hydrogenation and saved as PDBQT format files. Active pockets of these proteins were calculated using the Get Box plug-in PyMOL to facilitate the MD of the chemical components for Feikang prescription with the target site. A docking energy heat map related to components and COPD-related proteins was constructed by online platform. Final docking conformations underwent three-dimensional rendering visualization framework through Open-Source PyMOL (v2.5.0) (Giofrè et al., 2021).

MDs analysis

We started our investigation by building the topol file assigned with appropriate force. Subsequently, the protein-small molecule complexes were placed in a water box and the entire box was filled with the TIP3P water model to construct a solvent environment. After constructing the initial structure, we minimized the energy of entire system by the most rapid descent method for a total of 50,000 steps. Following the completion of the energy minimization process, we further equilibrated the system by raising the initial temperature to 310 K and the pressure to 1 bar. To prevent small molecules from moving violently in the initial stage of the temperature and pressure coupling process, we restricted their positions using a constrained molecular dynamics method.

After completing the aforementioned pre-equilibrium procedures, we removed all positional restrictions and proceeded to perform a conventional MDs with a simulation time of 100 nanoseconds. Respectively, VMD 1.9.3, PyMOL, Origin 2021 and LigPlus were utilized to view the trajectories, GROMACS's built-in tools were used to estimate the root-mean-square deviation (RMSD), root-mean-square fluctuations (RMSF) and radius of gyration (Rg). The solvent accessible surface area (SASA) was determined using GROMACS SASA utility. Qtgrace plugin was applied to analyze hydrogen bonds, defined as interactions between

two residues with a hydrogen donor. A hydrogen bond that maintained an occupancy of 50 % or higher during the simulation was considered stable. Additionally, the ligand-protein binding free energy landscape (FEL) and the eyelash plot were further estimated.

SPR analysis in vitro assay

The experiments of the binding affinities of pachymic acid with MPO, apigenin with p53 and MPO with p53 were performed at 25 °C on a BIAcore 1 K. To put it briefly, 200 μ M 1-ethyl-3-(3-dimethylamino-propyl) carbodiimide (EDC) and 50 μ M N-hydroxysuccinimide (NHS) were mixed and applied at 10 μ l/min for 420 s to activate a cell on the CM5 sensor chip. After combining 50 μ l of protein with 180 μ l of a pH 5.0, 10 mM sodium acetate solution, the protein was immobilized on the cell surface at a rate of 10 μ l/min for 420 s for two consecutive runs. And then, 1 M ethanolamine was utilized to block the cell. After that, PBS was used to equilibrate both aisles. Each run involved flowing the molecule stock solution at 10 μ l/min for 150 s after diluting it to a range of concentrations in PBS. Cells were regenerated for 5 min at the conclusion of each flow using a 10 mM glycine-HCl (pH=2.0) solution at a rate of 10 μ l/min. Biacore Insight (v.2.0, Cytiva) was used to gather data from the sample cell, which were then subtracted from those from the reference cell. Finally, BIAcore 1 K Evaluation software (Cytiva, Marlborough, MA, USA) was employed to globally fit the data to a 1:1 Langmuir binding model in order to determine the association and dissociation constants (Yi et al., 2021). The experimental data were exported to GraphPad Prism for generating corresponding figures.

In vivo experiments

Ethics statement and experimental animals

The Animal Experimental Ethical Committee of the First Affiliated Hospital of Guangzhou University of Chinese Medicine approved the study (Approval No.TCMF1–2021,079). We utilized SPF-grade, 6- to 8-week-old healthy male C57BL/6 J mice, weighing between 20 g and 25 g. The SPF-grade mouse feed and corn cob bedding were procured from Hunan Slack Guda Experimental Animal Co., Ltd. (Quality certification No.: SCXK(Hunan) 2019–0004). In compliance with the 3Rs of animal welfare, we aimed to minimize both the total number of animal usage and their distress. Ten animals were randomly allocated to each experimental group: control group, model group, low-dose Feikang prescription group, high-dose Feikang prescription group, and positive group. Mice were housed under a 12-h light/12-h dark cycle, with ambient temperature maintained at 22 ± 3 °C, humidity levels of 55 ± 15 %, and bedding changed once a week. The mice were provided with an appropriate amount of feed and autoclaved drinking water, and they were free to eat and drink as they pleased.

CS-induced COPD mice model and treatment

The continuous cigarette smoke exposure (CSE) method was utilized to generate a preclinical COPD model in C57BL/6 J mice. In this experiment, mice were put into a homemade smoke chamber, where 12 cigarettes (Tar: 11 mg, Nicotine: 0.9 mg, Carbon Monoxide: 12 mg/cigarette, Hongmei, China Hongta Tobacco Co., Ltd) were simultaneously ignited. The chamber was sealed, and the mice were exposed to smoke twice daily for 2 h per session, with intervals of more than 2 h between sessions. The mice were not exposed to smoke on Saturdays and Sundays, and were weighed every Saturday. This smoke exposure regimen was maintained from Monday to Friday for a total duration of 6 months. In both the control and model groups, saline (0.492 ml/30 g/d) was administered via intraperitoneal injection for 4 weeks, starting from the last day of the 6th month. Carbocysteine Oral Solution was purchased from the First Affiliated Hospital of Guangzhou University of Chinese Medicine (Guangzhou, China). The mint-flavored solution was available in 10 ml: 0.5 g sugar-free doses from Guangzhou Baiyunshan Pharmaceutical Holdings Co., Ltd. Carbocysteine was administered

intraperitoneally at a volume of 0.492 ml/30 g/d, corresponding to a dose of 1 mg/kg/d for 4 weeks. For Feikang prescription groups, the clinical equivalent high dose was administered at 1.64 mg/kg/d, whereas the low dose was half that, at 0.82 mg/kg/d.

Collection and counting of BALF cells

The left lung BALF was obtained by performing a transverse incision in the trachea and sequentially irrigating with 0.3 ml of normal saline three times. Then, the BALF was then centrifuged at 1000 rpm for 10 min at 4 °C. Following this, the pellet was washed with 500 µl of normal saline and spun it again. BALF was stained with Wright-Giemsa staining (Biosharp, #BL800A), and imaging was done with the Panoramic 250 Flash II Slide Scanner (3D HISTEC, Panoramic MIDI, Budapest, Hungary) at 200 × and 400 × magnification. The total number of cells in BALF was quantified employing the Countstar system (BioLab), and cell classification was carried out.

Lung histology

After fixing the right lung in 4 % paraformaldehyde (Biosharp, #70,081,800), we made 3 µm thick paraffin sections. Then, conventional Hematoxylin and eosin (HE) and Masson's trichrome staining protocols were implemented to assess inflammation cell infiltration, emphysema, and airway remodeling. Emphysema was assessed using Mean Linear Intercept (MLI) measurements: at a certain magnification, three spots of view were randomly picked out from each slice (avoiding large blood vessels and bronchioles). Then, we drew a cross in the middle of each field, and calculated the number of alveolar walls (Ns) that the cross went through and measured the total length (l) of the cross. MLI was calculated as $MLI = l/Ns$, indicating the average size of the alveoli. MLI measurements were performed independently by two investigators blinded to treatment groups to eliminate subjective bias. Results were averaged across three random fields per lung section avoiding vessels or bronchioles. The histological sections were scanned and analyzed at 200 × and 100 × magnification.

Cytokine and chemokine detection

We made use of ELISA to assess the levels of cytokine and chemokine proteins in serum. The serum levels of IL-6 (NeoBioscience, Cat#: EMC004), IL-8 (NeoBioscience, Cat#: EMC104.96), IL-1β (NeoBioscience, Cat#: EMC001b.96), TNF-α (NeoBioscience, Cat#: EMC102a), and MPO (MULTISCIENCES, Cat#: EK2133) were detected by ELISA following manufacturer's instructions. ELISA assays used $n = 6$ biological replicates per group. Each sample was analyzed in technical triplicate.

Prussian blue DAB staining

Stain the slides with DAB solution for approximately 5–10 min, monitoring the degree of color development microscopically. Dispose of the dye and wash the slides once with 0.01 mol/l PBS solution, followed by three washes with distilled water. Stain with hematoxylin for 1 min, rinse under running water, differentiate with dilute hydrochloric acid, and neutralize in ammonia solution before a final wash with tap water. Dehydrate sequentially in three changes of absolute ethanol (5 min each), clear in xylene (two changes, 5 min each), and mount with neutral balsam. Microscopically, iron deposits appear brown, nuclei stain light blue, and the background remains pale brown or clear. Images can be acquired and analyzed using a 3D HISTECH scanner.

Immunohistochemistry (IHC)

After deparaffinizing lung samples with xylene and ethanol, antigen retrieval was performed, followed by blocking with 3 % bovine serum albumin. They were then stained with mouse Wnt5a antibody (Bioss, Cat#: bs-1948R, 1:100), GPX4 antibody (Affinity, Cat#: DF6701, 1:50), p53 antibody (Affinity, Cat#: DF0879, 1:1000), and Heterochromatin Protein 1 (HP-1) antibody (Proteintech, Cat#: 11,831-1-AP, 1:50). Positive staining manifested as brown/tawny deposits, and three

representative fields per section were imaged using a 3D HISTECH scanner, with data expressed as Mean ± SEM.

IF and mIF

As advised by the manufacturer, the ROS immunostaining was carried out. The Reactive Oxygen species assay kit (Beyotime, Cat#: S0063) was used for reactive oxygen detection by H2DCFH-DA fluorescent probe. The paraffin sections were dewaxed in xylene and ethanol, followed by antigen retrieval, antibody incubation, and staining. After rinsing with running water, the sections were spun dry. Subsequently, multiplex imaging was conducted with a laser scanning confocal microscope (Leica TCS SPE II) to observe and analyze the fluorescence images with an oil-immersed × 40 objective. The primary antibodies employed were Wnt5a (Bioss, Cat#: bs-1948R, 1:100), GPX4 (Affinity, Cat#: DF6701, 1:50), and p53 (Affinity, Cat#: DF0879, 1:100), with Horseradish Peroxidase (HRP)-labeled goat anti-rabbit IgG (Servicebio, Cat#: GB23303, 1:500) used as secondary antibodies, in accordance with guidelines. Detection channels that had excitation at 377, 494, 543, and 628 nm as well as emission at 447, 527, 586, and 692 nm were applied for colocalization experiments (DAPI/SpGreen/SpOrange/Cy5).

Quantitative reverse transcription PCR (RT-qPCR) of lung tissues

Initially, lung tissue samples were homogenized, ensuring efficient cell lysis and release of RNA. Total RNA was extracted using TRIzol Reagent (Invitrogen™, Cat#: 15,596,026) in accordance with manufacturer's instructions. The RNA concentration and purity were assessed using a spectrophotometer. The RNA was then converted into cDNA utilizing a reverse transcription kit (Takara, Cat#: RR047A) and was analyzed by TB Green Premix EX Taq II (Takara, Cat#: RR820A). The PCR conditions consisted of 10 min of pre-denaturation at 95 °C, 15 seconds denaturation at 95 °C, and annealing at 60 °C for 30 seconds for 40 cycles. The mRNA expression levels were assessed using the comparative $2^{-\Delta\Delta CT}$ method, with normalization to β-actin as an internal control. The RT-qPCR method was utilized to validate the expression of the target genes. We also used the online tool BLAST (Basic Local Alignment Search Tool) in NCBI (<https://blast.ncbi.nlm.nih.gov/Blast.cgi>) for sequence alignment to verify the primer specificity. The primer sequences for p16, p21, p53, Ki-67, Wnt5a, GPX4, FTH-1, FTL, SLC7A11 and β-actin, provided by Tsingke Biotechnology Co., Ltd, Beijing, China, are listed in **Supplementary Table 1**.

WB

The proteins were denatured with SDS loading buffer, which were subsequently separated on 6 %, 10 %, and 12.5 % SDS-PAGE gels. Separated proteins were next transferred to PVDF membranes (Millipore, ISEQ00010) at 400 mA for 25 min in ice-cold transfer buffer with an electroporation buffer in a ratio of 1:2:7 using absolute ethanol and deionized water. The membranes were blocked with 5 % Skim milk (BD Difco™ Skim Milk, Cat#: 232,100) for 1 h at room temperature. The signal was visualized using ECL reagent (Affinity, Cat#: KF8005), and the intensity of the signals (IOD) was quantified using Image J (Fiji). The antibodies employed were as follows: p21 (Affinity, Cat#: DF6290, 1:1000), p53 (Affinity, Cat#: DF0879, 1:1000), Ki-67 (Affinity, Cat#: DF0198, 1:1000), Wnt5a (Bioss, Cat#: bs-1948R, 1:1000), GPX4 (Affinity, Cat#: DF6701, 1:1000), FTH-1 (Affinity, Cat#: DF6278, 1:1000), FTL (Proteintech, Cat#: 10,727-1-AP, 1:1000), and SLC7A11 (Affinity, Cat#: DF12509, 1:1000), Vinculin (HUABIO, Cat#: ET1705-94, 1:5000), HRP-conjugated β-actin (Proteintech, Cat#: HRP-6008, 1:8000), and Goat anti-Rabbit (Affinity, S0001, 1:3000).

Statistical analysis

The statistical analysis was calculated as Mean ± standard error of the mean (SEM) using GraphPad Prism (Version 9.5) software. The Shapiro-Wilk test was conducted to check for the normality of the data distribution, while homogeneity of variance was confirmed using

Bartlett test. One-way ANOVA was utilized to determine the differences between groups. For data comparisons between two groups, independent samples *t*-tests were performed. A *p*-value of <0.05 was deemed statistically significant.

Results

Identification of the constituents in Feikang prescription by UHPLC-MS/MS

As depicted in Fig. 1, the comprehensive analysis of the Feikang prescription by UHPLC-MS/MS yielded a total ion chromatogram (TIC). A total of 117 bioactive compounds including pachymic acid and apigenin were identified. (Supplementary Table 2). The potential active ingredients were entered into the TCMSP and Swiss TargetPrediction databases, yielding 761 targets.

Network pharmacology elucidated that Feikang prescription alleviated inflammation, oxidative stress, ferroptosis and cell senescence in COPD

The 117 chemical constituents and 761 corresponding targets of the Feikang prescription were imported into Cytoscape software to construct a “chemical constituents-targets” network (Fig. 2A). A total of 1454 COPD-related targets were identified after duplicate genes were eliminated. The 761 target genes of the Feikang prescription were intersected with the target genes associated with COPD, yielding 267 intersected genes identified as potential target genes (Fig. 2B). To view genes with close connection, the previously indicated 267 targets were ingested into the STRING database and then entered into the Cytoscape platform, screening out Hp, IL-6, MPO, IL-1 β , MMP9, TNF, and p53. These targets could serve as potential therapeutic targets for Feikang prescription against COPD, as shown in Fig. 2C-D.

GO functional and KEGG analysis of 267 intersecting targets was conducted using the PANTHER database. The biological process (BP) identified responses to positive regulation of GPX activity, regulation of GPX activity, positive regulation of oxidoreductase activity, epithelial cell apoptotic process, regulation of oxidoreductase activity, regulation of immune system process, programmed cell death, cell death, apoptotic process, cellular process. The relevant cellular component (CC) encompassed the PCNA-p21 complex and Wnt signalosome, among others. The molecular function (MF) identified including oxidoreductase activity and antioxidant activity. These processes and functions were all linked to Feikang prescription therapy for COPD (Fig. 2E). Additionally, KEGG enrichment analysis identified several signaling pathways as potential mechanisms for refractory COPD intervention, including p53 pathway feedback loops, p53 pathway, inflammation mediated by

chemokine and cytokine signaling pathway, oxidative stress response, and Wnt signaling pathway (Fig. 2F).

Inflammation, oxidative stress, programmed cell death, and GPX activity were closely related to ferroptosis. Besides, p21 and p53 were relevant to cell senescence, along with Wnt signalosome. The aforementioned results inspired us to propose the audacious hypothesis that the effect of Feikang prescription on treating COPD might be realized by regulating ferroptosis and cell senescence triggered by Wnt5a, a significant target in the Wnt pathway, and p53, a key target in the p53 pathway.

MD predicted initially the binding stability of MPO with pachymic acid, p53 with apigenin and MPO and p53

Based on the action pathway diagram and the degree values, seven main target genes were selected for subsequent MD, as detailed in Supplementary Table 3. MD was performed by ten primary components of the Feikang prescription with seven key targets (IL-6, IL-1 β , MPO, TNF, MMP9 and p53). The docking scores were visualized in a heat map (Fig. 3A). Through a careful analysis of the docking images, insights into the active compounds and target proteins were ascertained. Notably, the Hp with dihydroartemisinin, IL-1 β with apigenin, IL-6 with dihydroartemisinin, MMP9 with apigenin, MPO with pachymic acid, TNF with bryodulcosigenin, and p53 with apigenin were highlighted. Additionally, for five duplicate dockings with targets, we analyzed affinity trends of the primary active ingredients of Feikang prescription. A binding energy threshold of less than -9 kcal/mol elucidated that pachymic acid exhibited strong affinity with MPO. While the binding energy of less than -7 kcal/mol (Liu et al., 2022a) indicated that apigenin exhibited good binding activity with p53. Thus, p53 and MPO were selected out for further rigid-receptor MD (Fig. 3B-C). Supplementary Table 4 detailed the five docking results of protein-ligand interactions as well as the average inhibition constants. The binding mode of p53 and MPO was also predicted by eyelash plot (Fig. 3D). The results above demonstrated that MD predicts possible binding modes, but to determine whether ligands can bind stably, MDs provide further validation.

MDs indicated the binding stability of MPO with pachymic acid and p53 with apigenin

Analysis of RMSD results were revealed as the following: (1) The MPO-pachymic acid system was continuously oscillating within the first 10 ns, indicating that the system underwent structural rearrangements and adaptation upon ligand binding. (2) From 10 to 100 ns, the RMSD values plateaued and remained remarkably stable, suggesting that the

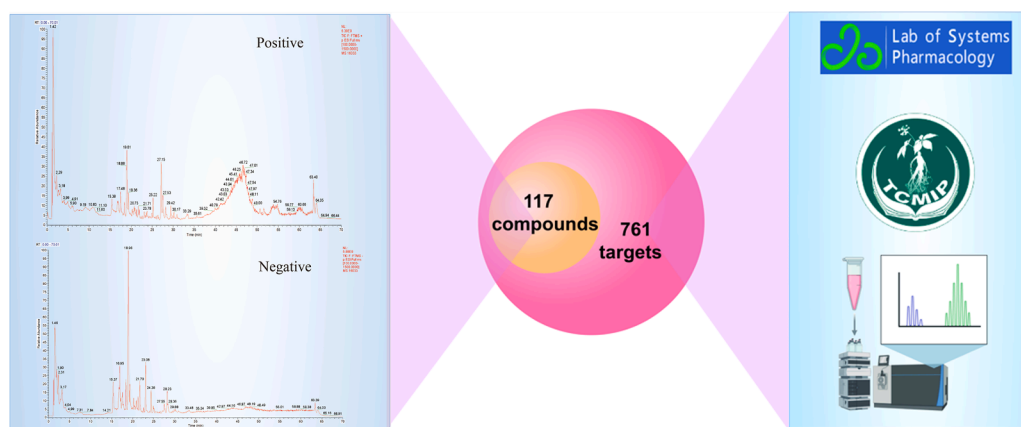


Fig. 1. Identification of the constituents in Feikang prescription by UHPLC-MS/MS. (Left: Positive ion mode, Negative ion mode; Medium: Numbers of chemical composition-target).

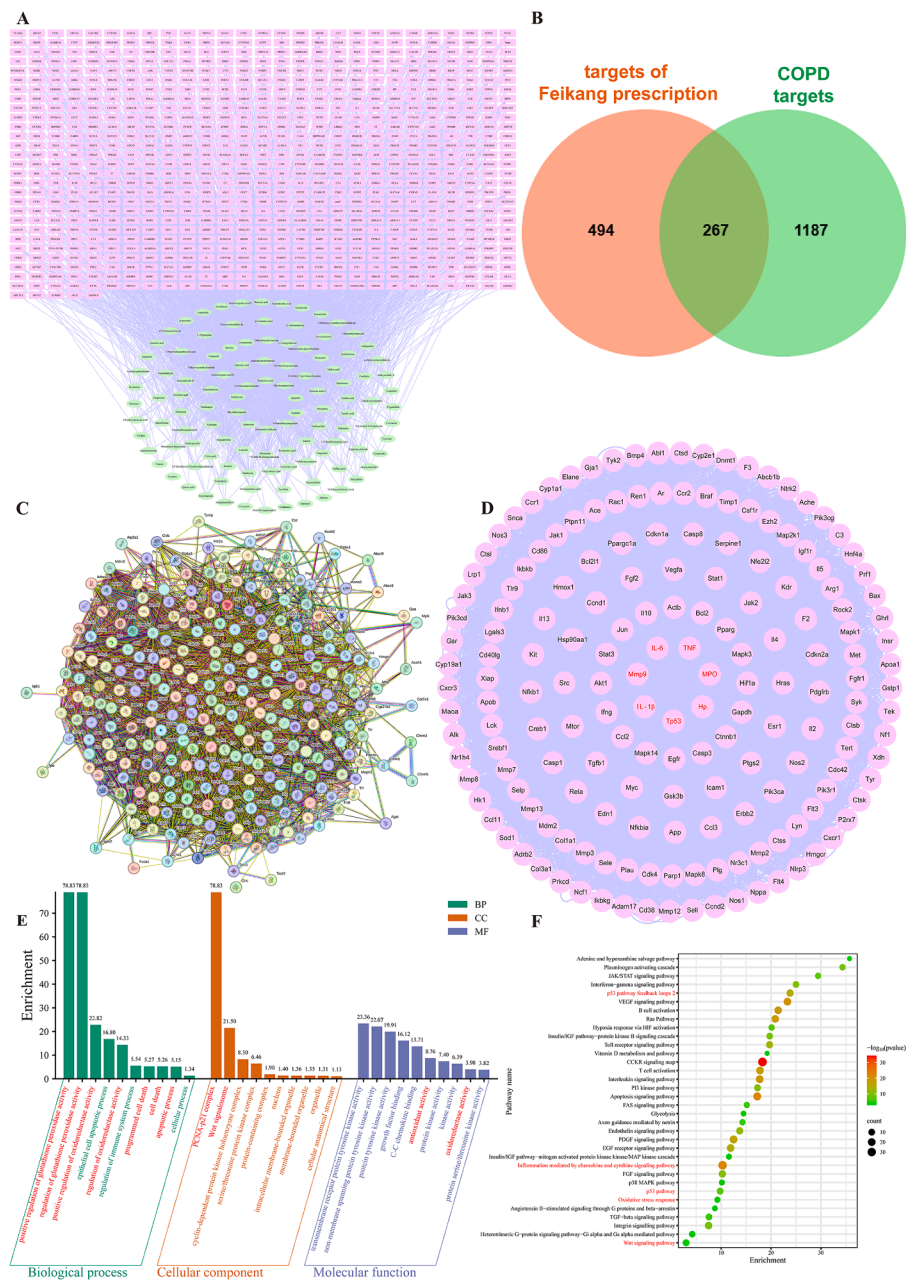


Fig. 2. Network pharmacology analysis. (A) Schematic diagram of the “chemical composition-target” network; (B) Venn diagram of potential targets of Feikang prescription for the treatment of COPD; (C) The interactive PPI network diagram of intersecting targets obtained from STRING database; (D) The 7 key target proteins that overlapped; (E) GO functional analysis; (F) KEGG pathway enrichment analysis.

MPO-pachymic acid complex achieved a stable binding conformation (Fig. 4A). To probe the flexibility of specific regions within the MPO protein, the RMSF was calculated for each residue. The target demonstrated significant conformational stability during the 100 ns MDs for all systems, according to the RMSF calculations, which were performed across the whole MDs trajectories with a few slight deviations at positions 400, 500, and 700 (Fig. 4B). In the meantime, we counted an average of 3 hydrogen bonds across the whole simulation (Fig. 4C). The contrast of eyelash plot of MPO with pachymic acid before and after MDs, indicating better position (Fig. 4D). The SASA was monitored to assess any significant changes in the overall protein conformation upon ligand binding. Our observations uncovered that the SASA values for the complexes were stable throughout the 100 ns MDs, implying no major alterations in the protein structure (Fig. 4E). The result unraveled a fairly consistent behavior (Fig. 4F). For the bound complex of MPO with

pachymic acid, frames at the minimal FEL were figured out. The FEL showed that the conformation corresponding to blue areas were selected as the lowest energy binding position (Fig. 4G). RMSD results for the p53 with apigenin complex showed that it remained in a stable state with minimal fluctuations around 0.25–0.3 nm (Fig. 4H). The RMSF result indicated that the target exhibited notable conformational stability with a few slight deviations at positions 25–180 (Fig. 4I). The simulation data indicated that the complex maintained an average of 1–2 hydrogen bonds across the whole simulation (Fig. 4J). Compared to the initial structure, the small molecule had a better pose and was well embedded inside the pocket (Fig. 4K). SASA values remained constant during the course of the 100 ns MDs, suggesting that the protein structure had not undergone any significant alterations (Fig. 4I). The Rg analysis also demonstrated a fairly consistent behavior throughout the MD simulation time, further supporting

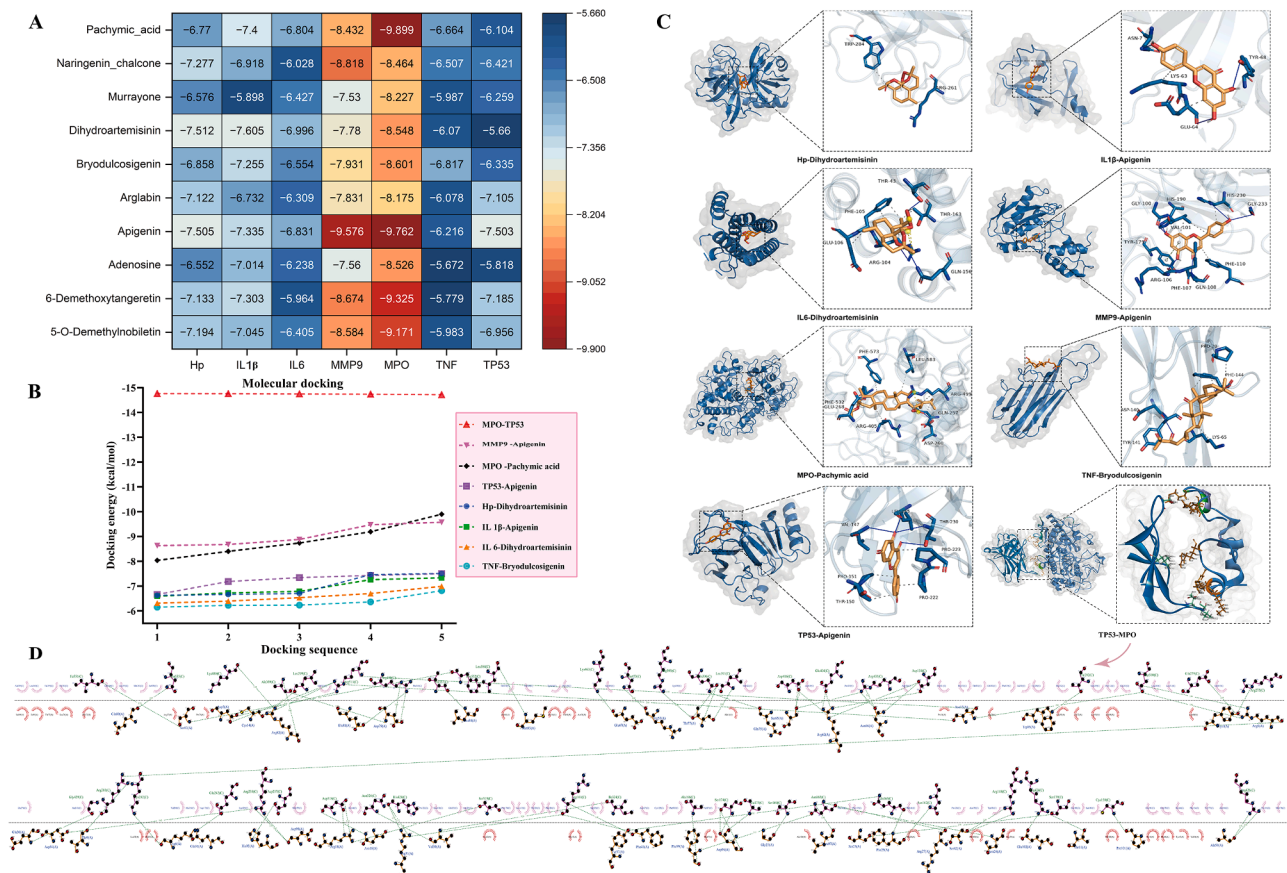


Fig. 3. MD assessment. (A) Heatmap of MD scores for chemical compositions and related targets. (B) Affinity trends of the main active constituents of Feikang prescription for 5 replicate dockings with targets; (C) The MD results of main active compounds of Feikang prescription and interaction between p53 and MPO; Hp-Dihydroartemisinin, IL-1β-Apigenin, IL-6-Dihydroartemisinin, MMP9-Apigenin, MPO-Pachymic acid, TNF-Bryodulcosigenin, p53-Apigenin, respectively; (D) The eyelash plot of interaction between p53 and MPO. In the binding model, hydrophobic interactions, hydrogens bonds, salt bridges were showed.

the overall stability of the p53-apigenin complex (Fig. 4M). For the bound complex of p53 and apigenin, frames at the minimal FEL were taken out (Fig. 4N). Collectively, MDs results supported the findings of MD that MPO and p53 can bind well with pachymic acid, apigenin.

SPR revealed the direct binding between pachymic acid with MPO, apigenin with p53 and MPO with p53

We conducted an SPR experiment *in vitro* to confirm the direct interaction of proteins with small molecule drugs and protein-protein. It showed that pachymic acid indeed exhibited a relatively strong binding affinity to MPO, with a KD value of 4.38 μM (Fig. 5A and B). Apigenin bounded to p53 with a KD of 1.74 μM (Fig. 5C and D). By contrast, the KD value of MPO with p53, which generated the highest affinity, was 0.0755 μM (Fig. 5E and F). Together, these results provided compelling evidence that the Feikang prescription's apigenin and pachymic acid target the key proteins p53 and MPO, respectively, through direct interactions, particularly MPO with p53.

Feikang prescription alleviated the inflammation and emphysema in the histopathological morphology of lung tissue in mice with CS-induced COPD

To clarify the molecular mechanism by which Feikang prescription reduced airway inflammation in COPD, we created a COPD model (Fig. 6A). HE staining results indicated that the COPD group exhibited significant inflammatory cell infiltration surrounding airways or blood vessels in lung tissues. Additionally, there was thickening of the alveolar walls, structural disorganization, and a marked increase in inflammation scores ($p < 0.0001$). In contrast, mice in both the low-dose and high-dose

Feikang prescription groups, as well as those in the carbocysteine group, demonstrated significantly reduced inflammatory cell infiltration around the bronchioles and blood vessels, with a more intact alveolar structure. The differences in inflammation scores were statistically significant ($p < 0.05$), as illustrated in Fig. 6B. Emphysema was evaluated using MLI measurements. The assessment involved calculating the total length of the cross line, the total number of alveoli within the field of view, and the number of alveolar intervals intersecting the cross line. The findings indicated that the MLI was significantly higher in the model group ($p < 0.01$). Furthermore, the MLI was significantly lower in both the low and high dose Feikang prescription groups when compared to the model group ($p < 0.01$) (Fig. 6C and F). Masson staining results indicated a significant increase in blue collagen deposition surrounding the airways and blood vessels in the model group ($p < 0.0001$). Administration of Feikang prescription and carbocysteine significantly ameliorated airway fibrosis (Fig. 6D and H). Taken together, the findings illustrated that Feikang prescription alleviated the inflammation and emphysema and fibrosis in the histopathological morphology of lung tissue in mice with CS-induced COPD.

Feikang prescription attenuated total and sorted BALF cell counts and inflammatory factor levels in COPD mice

Wright-Giemsa staining of BALF visualized and analyzed the distribution of various immune cell types present in the samples. This staining technique facilitated a detailed examination of the ratio of eosinophils, basophils, neutrophils, macrophages and lymphocyte within the groups (Fig. 7A). The analysis revealed an increase in the total number of cells present in BALF with model group (Fig. 7B). Meanwhile, cell

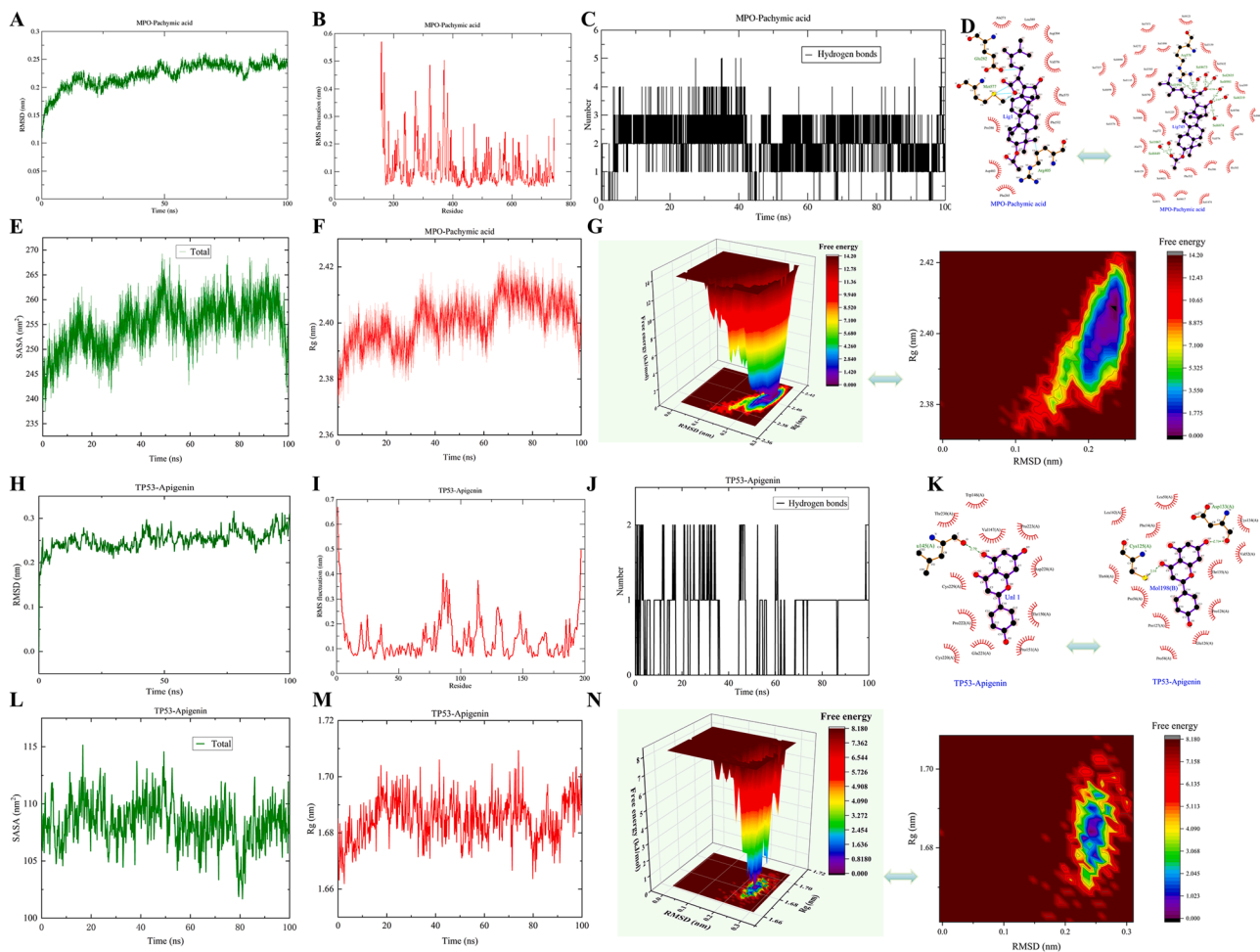


Fig. 4. Molecular dynamic simulations indicated the binding stability of MPO with pachymic acid and p53 with apigenin. (A) The RMSD trajectory analysis of bound complex of MPO with pachymic acid; (B) The RMSF analysis of MPO; (C) The Hydrogen bonds analysis of bound complex of MPO with pachymic acid; (D) The contrast of eyeflash plot of MPO with pachymic acid before and after molecular dynamic simulations (Left: before molecular dynamic simulations; Right: after molecular dynamic simulations); (E) The SASA analysis of bound complex of MPO with pachymic acid; (F) The Rg analysis of bound complex of MPO with pachymic acid; (G) Frames at the minimal FEL for the bound complex of MPO with pachymic (Left: 3D FEL; Right: 2D Gibbs Energy Landscape); (H) The RMSD trajectory analysis of bound complex of p53 with apigenin; (I) The RMSF analysis of p53; (J) The Hydrogen bonds analysis of bound complex of p53 with apigenin; (K) The contrast of eyeflash plot of p53 with apigenin before and after molecular dynamic simulations (Left: before molecular dynamic simulations; Right: after molecular dynamic simulations); (L) The SASA analysis of bound complex of p53 with apigenin; (M) The Rg analysis of bound complex of p53 with apigenin; (N) Frames at the minimal FEL for the bound complex of p53 with apigenin (Left: 3D FEL; Right: 2D Gibbs Energy Landscape).

classification indicated that there was a significant elevation in both neutrophils and eosinophils within the BALF in COPD group ($p < 0.05$). Accordingly, the percentage of macrophages was lower relative to that of neutrophils and eosinophils. The lymphocytes had no sense in BALF (Fig. 7C-F).

It is well established that inflammatory cytokines closely associated with airway inflammation following exposure to CS in COPD include IL-6, IL-8, IL-1 β , TNF- α , and MPO. Additionally, these cytokines were found to be elevated in the serum of the model group. Treatment with the Feikang prescription resulted in a reduction of these inflammatory cytokines (IL-6, IL-8, IL-1 β , TNF- α , and MPO) to varying degrees. Similarly, the carbocysteine group also demonstrated efficacy in alleviating the increase in these inflammatory cytokines. As anticipated, TNF- α , IL-1 β , IL-6 and IL-8 were all elevated in serum of COPD mice compared with control group ($p < 0.01$). Notably, the low-dose Feikang prescription group and high-dose Feikang prescription group suppressed the expression of TNF- α , IL-1 β , IL-6, IL-8 and MPO in serum samples ($p < 0.05$) (Fig. 7G-K). Carbocysteine effectively inhibited the expression of IL-6, IL-8, IL-1 β , TNF- α and MPO ($p < 0.01$). In sum, all above results provided compelling evidence that the Feikang prescription mitigated airway inflammation with CS-induced COPD mice.

Involvement of ferroptosis in COPD pathogenesis and modulation on the ferroptosis by Feikang prescription

Enhanced Prussian blue staining confirmed that there was a substantial increase in the deposition of iron ions within the lung tissues of COPD model mice, accompanied by ferroptosis. Feikang prescription could inhibit iron ion deposition. We found that compared with the control group, the nuclei of the cells in the lung tissue of model group appeared light blue, with a light brown or colorless background, indicating significant iron ion deposition. However, compared to the model group, the iron concentration was significantly lower following gavage with low-dose and high-dose Feikang prescription. The positive group exhibited less iron ion deposition compared with the COPD model group (Fig. 8A, D). ROS accumulation is regarded as a hallmark of ferroptosis (Peng et al., 2025; Zhu et al., 2021). In experiments, ROS levels in lung tissues was detected by ROS (DHE) staining of paraffin sections of mouse lung tissues using the ROS fluorescent probe DCFH-DA. The results showed that the level of ROS in the lung tissue of the model group was significantly increased, suggesting that oxidative stress activation is a major factor in COPD (Fig. 8B). It showed that Feikang prescription indeed reduces the ROS levels, and carbocysteine groups also inhibited

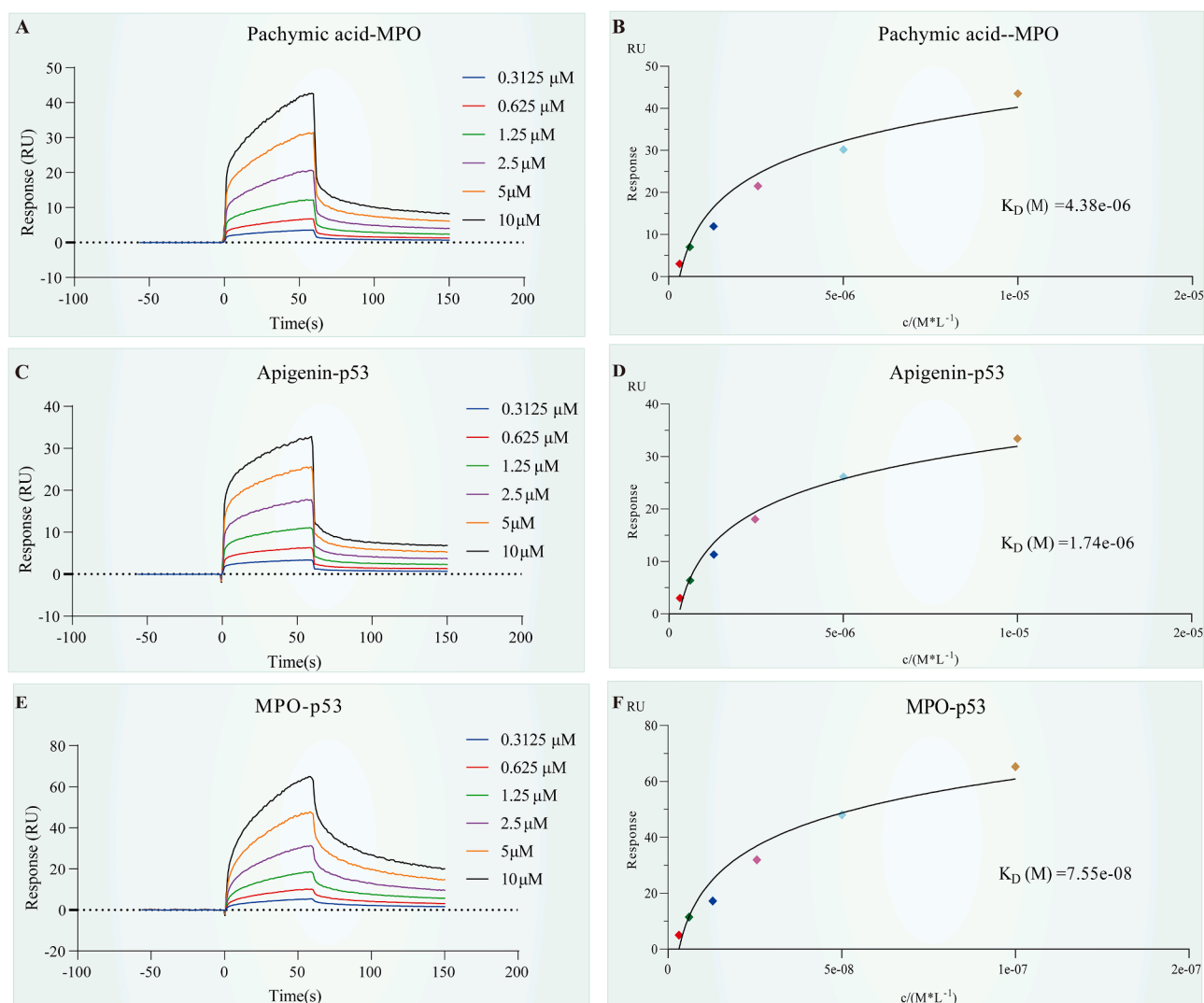


Fig. 5. The SPR analysis of pachymic acid with MPO, apigenin with p53 and MPO with p53. (A) *SPR* Sensorgrams at various increasing concentrations on pachymic acid with MPO; (B) The dissociation constant (K_D) obtained from the curve fitting; (C) *SPR* Sensorgrams at various increasing concentrations on apigenin with p53; (D) The dissociation constant (K_D) obtained from the curve fitting; (E) *SPR* Sensorgrams at various increasing concentrations on MPO with p53; (F) The dissociation constant (K_D) obtained from the curve fitting.

the ROS levels (Fig. 8E).

To explore the molecular mechanism and key targets related to ferroptosis in COPD, GPX4, FTH-1, FTL and SLC7A11 were identified as key genes. We further detected the mRNA expression levels of GPX4, FTH-1, FTL, SLC7A11, which were downregulated in COPD group ($p < 0.05$). After treatment with low-dose Feikang prescription, high-dose Feikang prescription and carbocysteine, the mRNA expression levels of these genes were upregulated compared to the model group, except for SLC7A11 with no statistical significance in the low-dose Feikang prescription group (Fig. 8F-I). And the melting curves of the samples were illustrated for the GPX4, FTH-1, FTL, SLC7A11 assay, confirming primer specificity with single peak at $T_m \approx 83-84, 84-86, 88-90$ and $85-86^\circ\text{C}$, respectively (Fig. S1A-D). Meanwhile, the protein levels of GPX4, FTH-1, FTL and SLC7A11 in lung tissue were also quantified by WB (Fig. 8C). WB demonstrated that GPX4 expression was significantly lower in model group. Furthermore, FTH-1, FTL and SLC7A11 expression levels were decreased, correlated with ferroptosis in COPD. After treatment with low-dose and high-dose Feikang prescription, as well as the carbocysteine, the expression levels of these proteins were upregulated compared to the model group (Fig. 8J-M).

Involvement of cellular senescence in COPD and modulation by Feikang prescription

RT-qPCR validation was performed on genes associated with cellular senescence, such as p16, p21, p53 and Ki-67. In comparison to the control group, the expressions of p16, p21, p53 and Wnt5a were elevated in the model group ($p < 0.01$). Conversely, a significant decrease in Ki-67 was observed in the model group ($p < 0.01$). Following the administration of both low-dose and high-dose Feikang prescription, as well as the treatment with carbocysteine, there was a notable reduction in the mRNA expression levels of p16, p21, and p53. After low-dose Feikang prescription, high-dose Feikang prescription and carbocysteine treatments, the mRNA expression levels of cellular senescence-related genes (p16, p21 and p53) were significantly decreased ($p < 0.05$) (Fig. 9A-C). Otherwise, the Ki-67 was expressed lowly in the model group but was significantly increased after treatment with low-dose, high-dose of Feikang prescription, and carbocysteine (Fig. 9D). Notably, Wnt5a, a gene mediating cellular senescence, was highly expressed in the COPD group and significantly downregulated in Feikang prescription group and carbocysteine group, which was consistent with our expected results (Fig. 9E). Also, the melting curves of

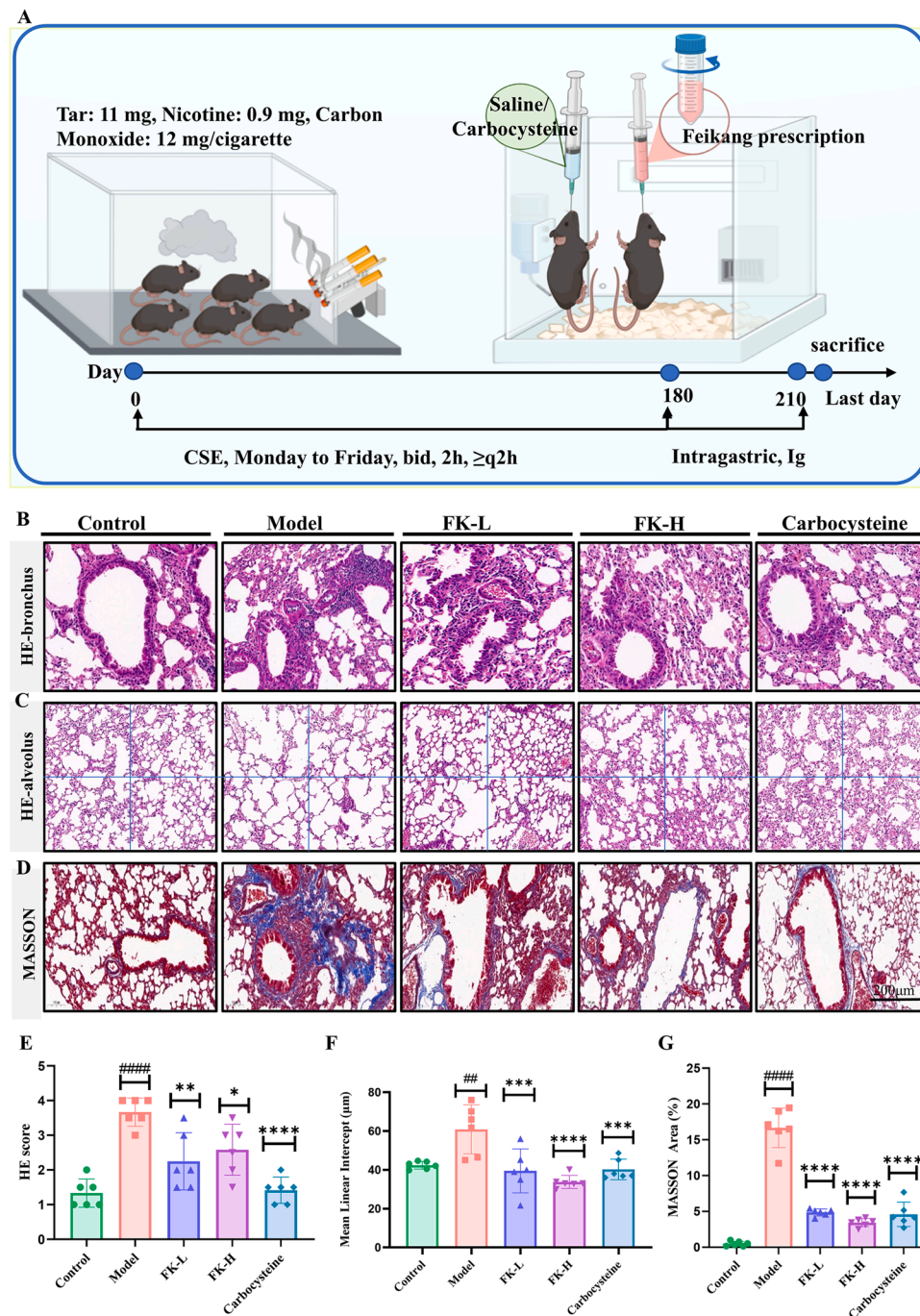


Fig. 6. Construction of a CS-induced COPD mice model. (A) Flowchart of mouse model establishment; (B–D) HE staining for airway inflammation, pulmonary emphysema, and Masson's trichrome staining of the lungs of control, model, low-dose and high-dose Feikang prescription group, and Carbocysteine group. (20x original magnification, scale bar: 200 μm); (E) Statistics analysis of HE scores between the groups; (F) Statistics analysis of mean linear intercept between the groups (μm); (G) The area of collagen fibres of Masson's trichrome staining (N = 6); Control group vs model group, $^{\#}p < 0.05$, $^{\#\#}p < 0.01$, $^{\#\#\#}p < 0.001$, $^{\#\#\#\#}p < 0.0001$; FK-L, FK-H, Carbocysteine group vs model group, $^*p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$; $^{****}p < 0.0001$. FK-L: low-dose Feikang prescription group; FK-H: high-dose Feikang prescription group.

the p16, p21, p53, Ki-67, Wnt5a and actin, confirmed primer specificity with single peak at $T_m \approx 88-89$, $86-87$, $87-88$, $80-81$, $83-84$ and $88-89$ °C correspondingly (Fig. S1E–J). Additionally, the protein levels of p21, p53 and Wnt5a were upregulated, quantified by WB ($p < 0.05$). WB demonstrated that Ki-67 expression was significantly lower in model groups than in control group ($p < 0.01$). After therapy with low-dose, high-dose Feikang prescription and carbocysteine, the expression levels of p21, p53, and Wnt5a were suppressed compared to the model group, except for Ki-67 (Fig. 9F–J).

Feikang prescription regulated GPX4/Wnt5a/p53 signaling pathway to attenuate airway injury in COPD

IHC staining was exhibited to validate the expression and localization of GPX4, Wnt5a, p53 and HP-1. HP-1 played a key role in cellular senescence. The IHC staining results for GPX4 further confirmed the presence of ferroptosis in the COPD mice model. Compared with the control group, GPX4 protein expression was suppressed in the lung tissues of mice in the model group, while the expression of Wnt5a, p53, and

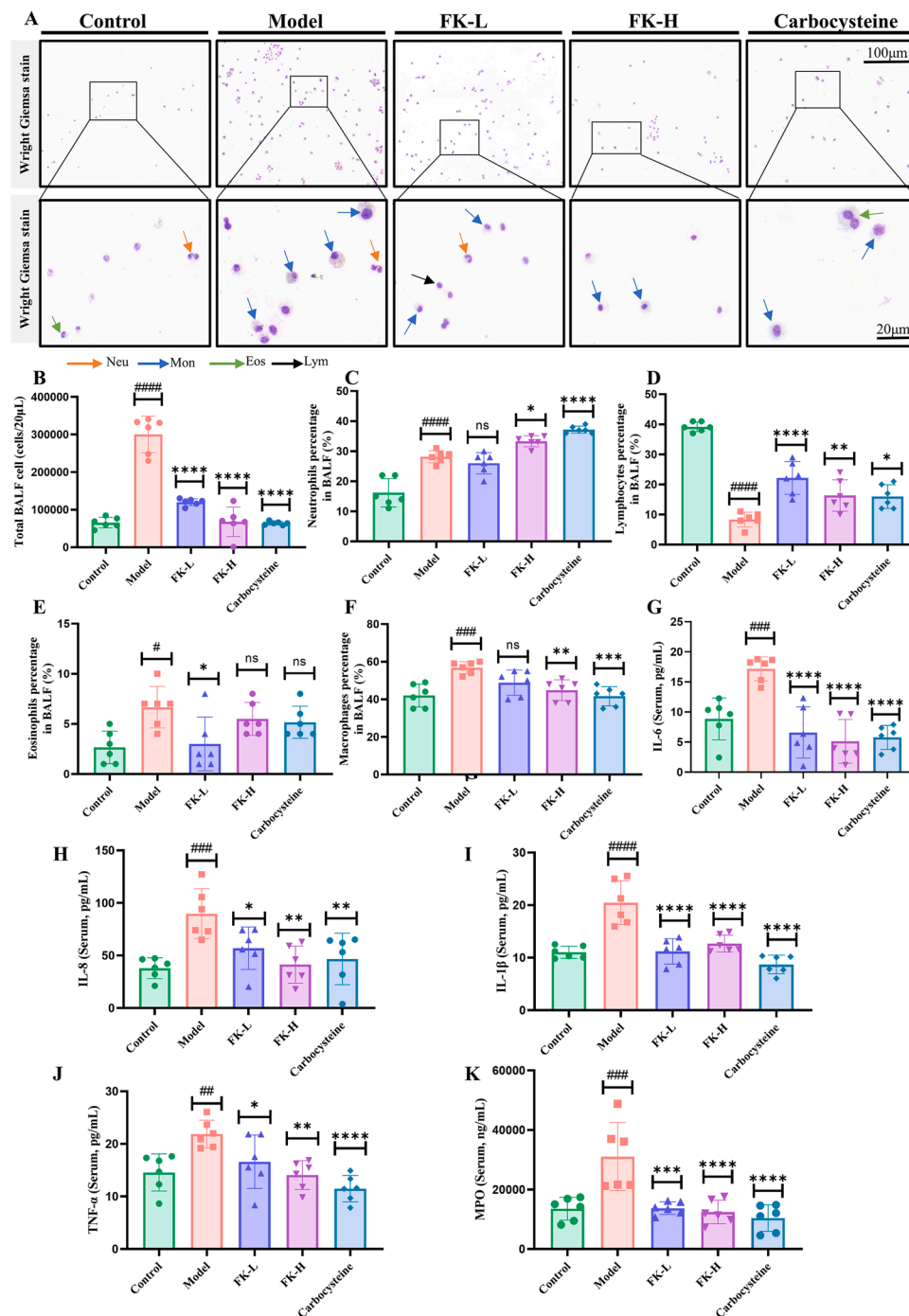


Fig. 7. Airway inflammation of a CSE-induced COPD mice model. (A) BALF cells smears. (20x original magnification, scale bar: 200 μm); (B-F) Total cells counts and four different cells counts in BALF respectively; (G) IL-6, (H) IL-8, (I) IL-1β, (J) TNF-α and (K) MPO levels in serum were measured by ELISA (N = 6); Control group vs model group, #p < 0.05, ##p < 0.01, ###p < 0.001, ####p < 0.0001; FK-L, FK-H, Carbocysteine group vs model group, *p < 0.05, **p < 0.01, ***p < 0.001, ****p < 0.0001. FK-L: low-dose Feikang prescription group; FK-H: high-dose Feikang prescription group.

HP-1 was significantly increased. Nevertheless, the protein levels of Wnt5a, p53, and HP-1 in the lung tissues dropped after therapy of Feikang prescription ($p < 0.001$), indicating that cellular senescence existed in the COPD mice model, as shown in Fig. 10A, B-E.

Co-Localization of Wnt5a, GPX4 and p53 alleviated airway injury in COPD mice

To further investigate whether the GPX4-dependent ferroptosis is related to the Wnt5a/p53-mediated cellular senescence in COPD, we

identified and quantified the expression of GPX4, Wnt5a and p53 (red, green, and yellow fluorescence). The mIF showed co-localization of GPX4, Wnt5a and p53 in the COPD mice. The IF in lung sections of COPD mice further testified increased distribution of Wnt5a and p53, whereas Feikang prescription downregulated the expression levels of these two crucial proteins, and the carbocysteine group showed similar effects. Furthermore, IF staining of lung tissues from COPD mice revealed decreased distribution of GPX4. However, treatment with the Feikang prescription elevated the expression levels of these key proteins, and the carbocysteine group upregulated its expression levels compared with

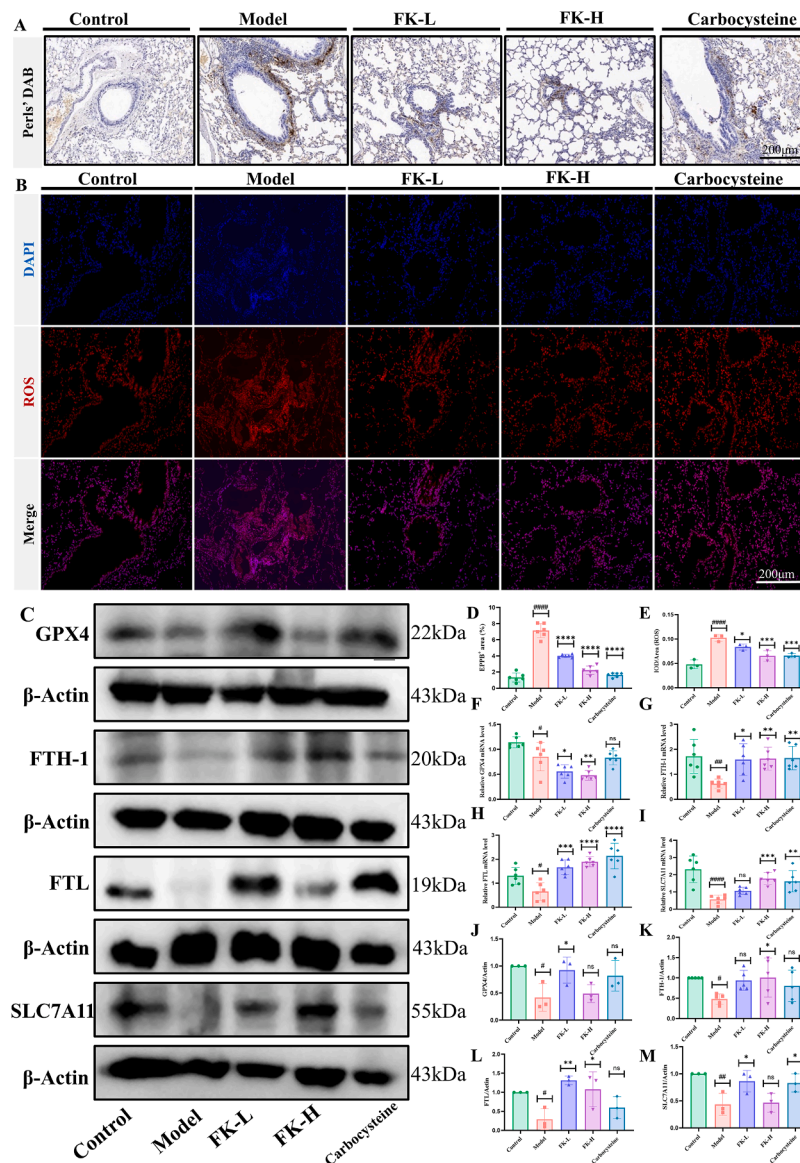


Fig. 8. Involvement of ferroptosis in COPD pathogenesis and modulation on the ferroptosis by Feikang prescription. (A) Perls' DAB; (B) IF detection of ROS level in mice lung tissues (20x original magnification, scale bar: 200 μ m); (D) Statistics analysis of Perls' DAB between the groups ($N = 6$). (E) Statistical plots of ROS expression in lung tissues of control, model, Feikang prescription groups and Carbocysteine groups ($N = 3$). (F-I) RT-qPCR analysis about the mRNA expression levels of GPX4, FTH-1, FTL, SLC7A11 with the method of $\Delta\Delta Ct$ ($N = 6$). (C, J-M) WB showing expression levels of GPX4, FTL, FTH-1 and SLC7A11 and β -actin in COPD ($N = 3$). Control group vs model group, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$. FK-L: low-dose Feikang prescription group; FK-H: high-dose Feikang prescription group.

the model groups. GPX4 and Wnt5a, Wnt5a and p53, and GPX4 and p53 were all closely linked by the colocalization curves (Fig. 11A-G). Therefore, we postulated that GPX4/Wnt5a/p53 signaling axis might be involved in COPD pathogenesis and modulated by Feikang prescription.

Discussion

The pathogenesis of COPD remains highly complex, characterized by a multifaceted interplay of chronic airway inflammation, oxidative stress, as well as the alveolar structure damage. The drawbacks of current pharmacotherapies, which often feature limited target specificity, restricted clinical utility, and suboptimal medication adherence, hinder the establishment of standardized treatment strategies for COPD (Xu et al., 2025). Glucocorticoids and bronchodilators are typically effective in modulating inflammatory responses (Wang et al., 2020). However, individuals diagnosed with COPD frequently demonstrate a diminished responsiveness to ICS, thereby constraining the spectrum of viable

therapeutic interventions. As a result, TCM has garnered escalating acknowledgment as a therapeutic modality for COPD, like the Feikang prescription (Yang et al., 2020), is particular in its multi-constituent composition, multi-target engagement, and multi-pathway therapeutic benefits. Our investigation found Feikang prescription attenuated airway inflammation, airway remodeling and regulated the “airway inflammation- ROS production and oxidative stress-ferroptosis and cell senescence” axis by activating the expression of GPX4 and inhibiting ROS, Wnt5a, p53 expression as well as oxidative stress response, p53 pathway and Wnt signaling pathway.

Our study showed that apigenin, dihydroartemisinin, pachymic acid, bryodulcosigenin were the critical bioactive constituents of Feikang prescription in the treatment of COPD. Apigenin, in particular, exhibits anti-inflammatory, enhancing oxidative stress resistance, and anti-aging effects (Gu et al., 2025; Sun et al., 2025). Pachymic acid reduces oxidative stress damage (Zhang et al., 2021). Network pharmacology analysis was performed to uncover the targets and mechanisms of

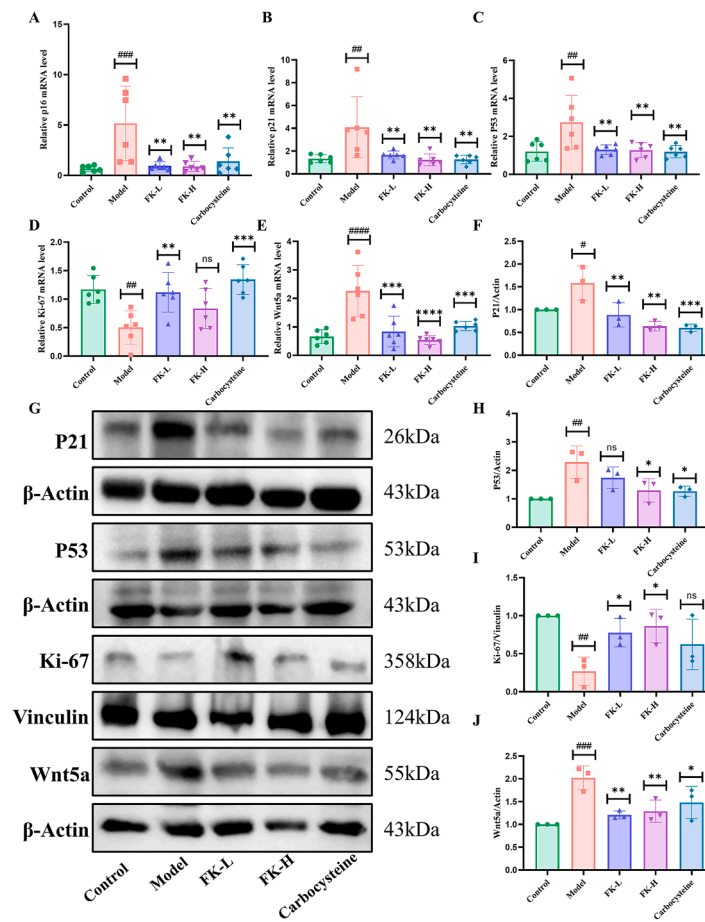


Fig. 9. Involvement of cellular senescence in COPD and modulation by Feikang prescription. (A-E) RT-qPCR analysis about the mRNA expression levels of p16, p21, p53, Ki-67 and Wnt5a with the method of $\Delta\Delta Ct$ ($N = 6$). (F-J) WB showing expression levels of p21, p53, Wnt5a and β -actin as well as Ki-67 and Vinculin in COPD ($N = 3$). Control group vs model group, $^{\#}p < 0.05$, $^{\#\#}p < 0.01$, $^{\#\#\#}p < 0.001$, $^{\#\#\#\#}p < 0.0001$; FK-L, FK-H, Carbocysteine group vs model group, $^*p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$; $^{****}p < 0.0001$. FK-L: low-dose Feikang prescription group; FK-H: high-dose Feikang prescription group.

Feikang prescription in treating COPD. These findings disclosed 117 targets overlapping the disease genes of COPD. Among them, 7 related targets were identified for enrichment analysis based on PPI network analysis results, including pro-inflammatory cytokines (IL-6, IL-1 β , MPO, TNF & MMP9), the related gene of oxidative stress (MPO) (Shao et al., 2020), relevant target of ferroptosis (Serotransferrin, Hp) (Feng et al., 2024) and core target of cellular senescence (p53) (Miller et al., 2025). The BP identified include responses to positive regulation of GPX activity, regulation of GPX activity, positive regulation of oxidoreductase activity, epithelial cell apoptotic process, regulation of oxidoreductase activity, regulation of immune system process, programmed cell death, cell death, apoptotic process, and cellular process. The relevant CC encompassed the PCNA-p21 complex and Wnt signalosome. The MF identified include oxidoreductase activity and antioxidant activity. The KEGG enrichment analysis revealed that the relevant targets were primarily enriched in p53 pathway by glucose deprivation, p53 pathway, Oxidative stress response, Inflammation mediated by chemokine and cytokine signaling pathway, and Wnt signaling pathway. Network pharmacology results provide preliminary evidence that Feikang prescription may treat COPD by modulating inflammatory reaction, oxidative stress, regulation of GPX activity and cellular senescence. Inflammatory reaction, oxidative stress, regulation of GPX activity are closely linked to ferroptosis. Relevant research highlights a dynamic interplay among inflammation, oxidative stress, ferroptosis, and cellular senescence in disease pathogenesis (Lu et al., 2025; Zhou et al., 2025a).

Afterwards, MDs was employed to evaluate the binding interactions between bioactive compounds and core target proteins. The analysis

revealed that p53 exhibited a strong binding affinity toward the primary target MPO, with a binding energy of -11.28 kcal/mol. This finding supports the link between oxidative stress and cellular senescence. Oxidative stress contributes to increased intracellular iron content, resulting in accumulation of lipid peroxides and activation of ferroptosis (Zhang et al., 2024a). This further provides evidence of the correlation between ferroptosis and cellular senescence. Analysis of MDs parameters in RMSD, RMSF, SASA, Rg, and FEL revealed that Feikang prescription may exert its therapeutic effects by stabilizing the binding of MPO with pachymic acid and p53 with apigenin. The results implied a possible association between the Wnt5a/p53-mediated cellular senescence and the GPX4-ferroptosis in individuals with COPD. In previous studies, MDs had been applied to verify TNF and IL-6 with other bioactive compounds in COPD (Pang et al., 2024). The direct binding of apigenin with p53, pachymic acid with MPO, and MPO with p53 was discovered using the SPR technique, which contributes to a better understanding of the underlying molecular pathways. Hence, integrated network pharmacology, MD, MDs and SPR analysis, GPX4, FTH-1, FTL, SLC7A11 and SASP, p16, p21, p53, Ki-67 were identified as the key targets related to the mechanisms of ferroptosis, cellular senescence, respectively (Wang et al., 2024, 2025b; Zhou et al., 2024).

Subsequently, a COPD mouse model was established through CSE, followed by the validation of COPD airway inflammation and emphysema using HE staining. Masson staining was employed to reflect collagen fiber deposition in the airways, while the levels of inflammation were assessed through BALF total and sorted cell counts, along with ELISA assays, which further illustrated the airway injury associated with

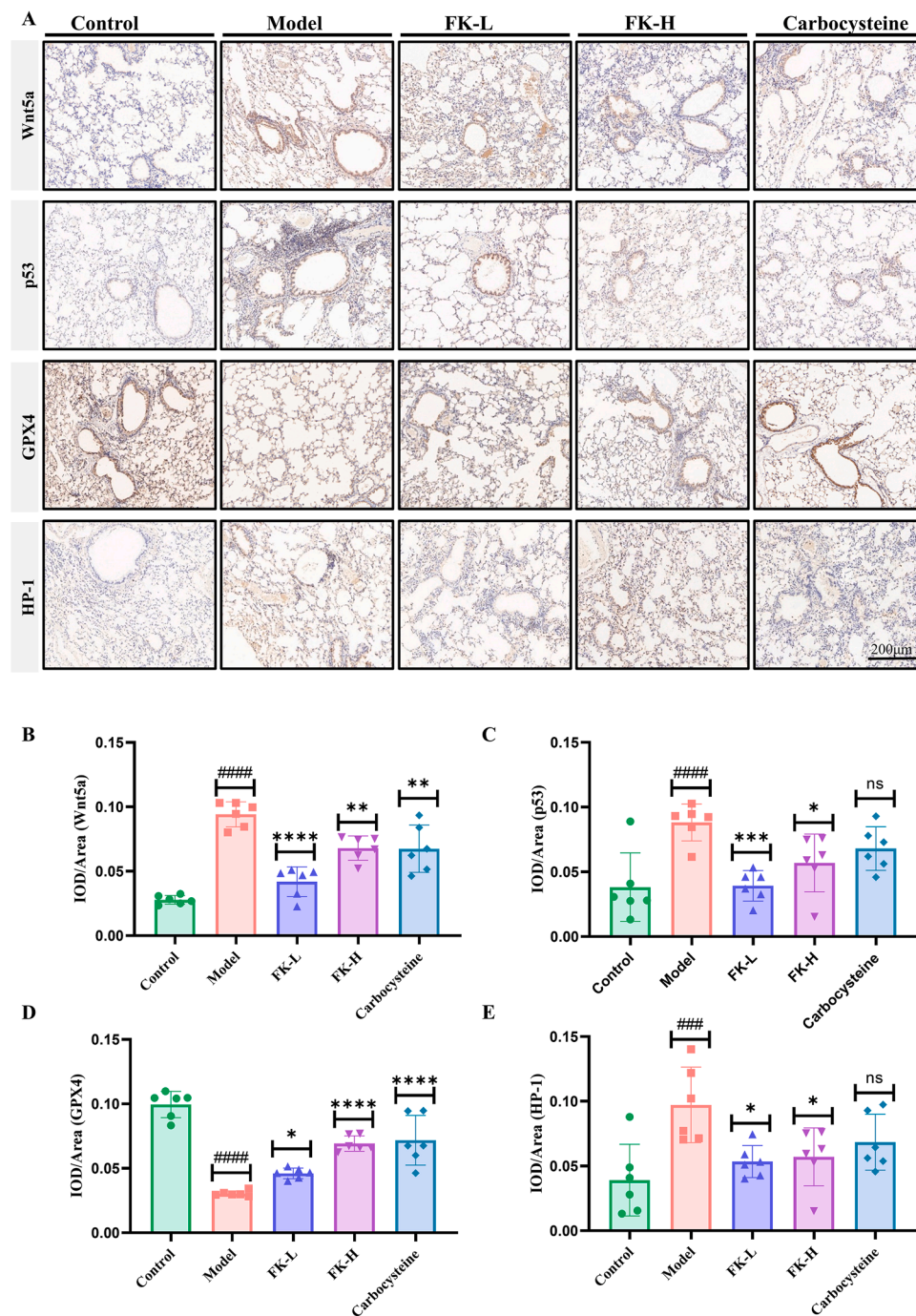


Fig. 10. Localization and expression level of GPX4, Wnt5a, p53 and HP-1 by IHC. IHC staining of (A) Wnt5a, p53, GPX4, and HP-1 expression in lung tissues of control, model, Feikang prescription groups and Carbocysteine groups (20x original magnification, scale bar: 200 μm). (B-E) Statistics plots of GPX4, Wnt5a, p53 and HP-1 by IHC between the groups ($N = 6$). Control group vs model group, $^{\#}p < 0.05$, $^{\#\#}p < 0.01$, $^{\#\#\#}p < 0.001$, $^{\#\#\#\#}p < 0.0001$; FK-L, FK-H, Carbocysteine group vs model group, $^*p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$; $^{****}p < 0.0001$. FK-L: low-dose Feikang prescription group; FK-H: high-dose Feikang prescription group.

COPD. Next, GPX4-dependent ferroptosis was verified by Prussian blue staining, qPCR and WB, and the activation of oxidative stress level in COPD mice was demonstrated by IF. Both oxidative stress and cellular senescence are featured by an increase in ROS production (Suryadevara et al., 2024; Xiaofei et al., 2022). Airway inflammation, MPO, ROS production and oxidative stress may result in ferroptosis or cellular senescence. The findings on the GPX4-ferroptosis of COPD align with previous investigations by Yoshida et al. (2019), who either confirms that GPX4 expression is decreased in COPD patients and further shows that iron accumulation and lipid peroxidation are associated with ferroptosis, providing a new understanding of COPD pathogenesis.

Furthermore, smoking can lead to iron accumulation and promote lipid peroxidation, which correlates with GPX4 activity, thereby highlighting the role of CSE in trigger and formation of ferroptosis (Li and Wang, 2022). There is mounting evidence suggests that ferroptosis is intricately linked to inflammation, and oxidative stress. The oxidative system consists of iron ions and ROS, which counterbalances the antioxidant system composed of the cystine/glutamate antiporter (System Xc⁻) and GPX4 (Zhang et al., 2024b). Whereas oxidative stress in COPD also promotes lipid peroxidation and iron accumulation, driving ferroptosis. In addition, increasing researches indicates 7-dehydrocholesterol (7-DHC) modulates ferroptosis (Cui and Ye, 2024; Freitas et al., 2024;

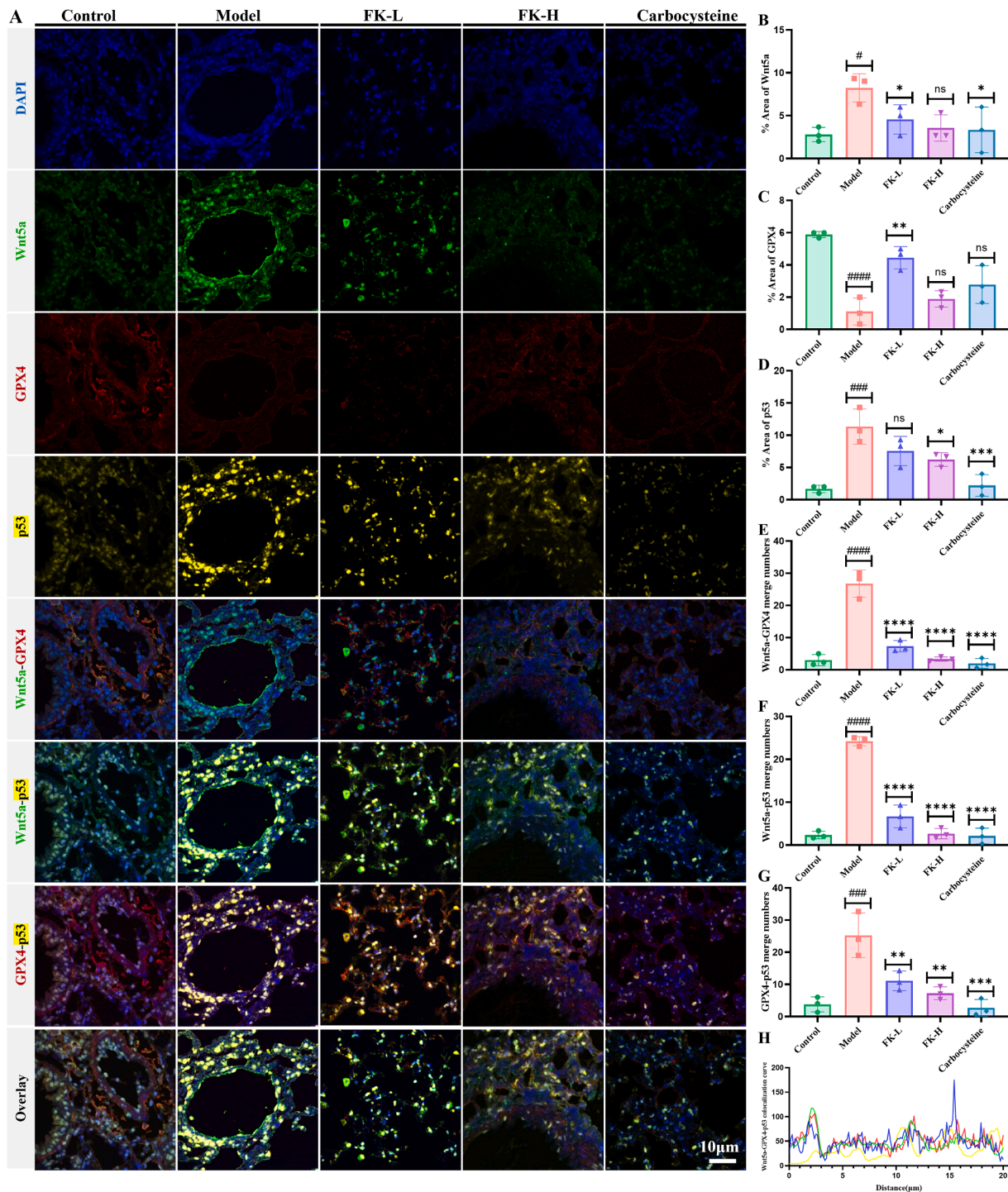


Fig. 11. Effects of Feikang prescription on co-localization of Wnt5a, GPX4 and p53 with COPD mice. (A) Co-localization of Wnt5a, GPX4 and p53 ($\times 400$); (B) Statistics analysis on %area of Wnt5a; (C) Statistics analysis on %area of GPX4; (D) Statistics analysis on %area of p53; (E) The colocalization counts of GPX4-Wnt5a; (F) The colocalization counts of GPX4-p53; (G) The colocalization counts of Wnt5a-p53; (H) The colocalization curve of Wnt5a, GPX4, p53 and DAPI. (40x original magnification, scale bar: 10 μ m). The data were expressed as the Mean $\pm$ SEM ($N = 6$). Control group vs model group, # $p < 0.05$, ## $p < 0.01$, ### $p < 0.01$, #### $p < 0.0001$; FK-L, FK-H, Carbocysteine group vs model group, * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; **** $p < 0.0001$. FK-L: low-dose Feikang prescription group; FK-H: high-dose Feikang prescription group.

Li et al., 2024). Lee et al. predominantly explores the intricate association between ferroptosis and cholesterol metabolism pathways, mainly focusing on the cholesterol biosynthesis pathway (Lee and Roh, 2024). Numerous studies have shown that ferroptosis is linked to cholesterol metabolism, with cholesterol sometimes acting to suppress ferroptosis (Liu et al., 2023; Sun et al., 2023, 2022). Specifically, the mevalonate pathway, involved in cholesterol synthesis, produces intermediates like

isopentenyl pyrophosphate (IPP), which are important for the function of proteins like GPX4 that protect cells from ferroptosis (Liu et al., 2023; Sun et al., 2023, 2022). Squalene Epoxidase (SQLE) is a key enzyme involved in cholesterol biosynthesis, and its degradation can increase levels of CoQ10 and squalene, which can protect cells from ferroptosis (Liu et al., 2023; Sun et al., 2023, 2022). The interplay between cholesterol metabolism and ferroptosis has implications for

understanding diseases and for developing potential therapeutic strategies. Although the current study did not perform direct experimental assays related to cholesterol metabolism, we hypothesize a potential indirect association between Feikang prescription's ferroptosis-modulating effects and cholesterol-related pathways. This assumption is based on the observed suppression of lipid ROS and restoration of GPX4 levels, both of which are downstream events potentially influenced by sterol metabolism. Future research is warranted to further explore whether Feikang prescription modulates ferroptosis through regulating cholesterol metabolism.

The occurrence of bronchial epithelial cell senescence linked to Wnt5a (a typical ligand in the Wnt signaling pathway) (Mastelaro de Rezende et al., 2020), was confirmed in the COPD model mice through qPCR and WB. There are two types of Wnt signaling pathways: the canonical pathway and noncanonical pathway. The promotion of cell division, maturation, and proliferation is greatly aided by the canonical route (Liu et al., 2022b). The direct recruitment of Dishevelled and the activation of Dvl proteins enables the Wnt5a's binding to Fzd upon Wnt signaling activation. For noncanonical pathways, this step then triggers the transcription of downstream target genes, which increases cell proliferation and resistance to apoptosis (Bowin et al., 2023). Related studies have shown that CS exposure induces lung cell senescence in p16-3MR mice and the removal of these senescent cells, which are induced by CS, has been found to reverse the influx of neutrophils in young groups. The elimination subsequently leads to the restoration with characteristic of the upregulated genes associated with mitochondrial function in p16-3MR mice, highlighting a significant interaction between senescence, inflammation, and cellular health in the context of lung disease (Kaur et al., 2023). Weihong Chen's team demonstrated that the non-classical WNT signaling pathway conveyed by Wnt5a, plays a significant role in inflammatory responses and epithelial-mesenchymal transition induced by benzo(a)pyrene. Further investigation into the underlying mechanisms revealed that the HSA-122-5P/TP53 axis regulates the non-classical Wnt signaling pathway mediated by WNT5A, which subsequently modulates benzo(a)pyrene-induced damage to airway epithelial cells (Fan et al., 2023). Oxidative stress caused by tobacco smoke promotes EMT in bronchial epithelial cells, thereby activating signaling pathways, such as the Wnt family of wingless MMTV integration site members and transforming growth factor beta (TGF- β), along with other signaling pathways, leading to airway remodeling in COPD (Zhou et al., 2023). This investigation aimed to dissect the relationship between GPX4-dependent ferroptosis and Wnt5a/p53-mediated cellular senescence. Additionally, the localization and expression levels of GPX4, Wnt5a, p53, and HP-1 were assessed via IHC, and mIF was utilized to elucidate the crosstalk among GPX4, Wnt5a and p53. This approach aimed to demonstrate that therapeutic effects of Feikang prescription involve modulation between GPX4-dependent ferroptosis with Wnt5a/p53-mediated senescence in bronchial epithelial cells. Rodencal et al. (2024) report that p53 expression attenuates ferroptosis sensitivity, while direct inhibition of GPX4 augments this sensitivity. This study investigates the role of ferroptosis in p53-mediated tumor suppression, focusing on the regulatory pathways by which p53 influences both classical (GPX4-dependent) and non-classical (GPX4-independent) ferroptosis pathways, as well as the critical cytokines that contributed to p53-dependent ferroptosis during tumor suppression (Liu and Gu, 2022). Additionally, it has been observed that GPX4 expression can be changed by transcription factors like p53 (Fan et al., 2025). Ferroptosis and cellular senescence have been shown to occur together, and ferroptosis inhibitors greatly slow down the development of cell senescence, whereas inducers accelerate it (Fu et al., 2025). To reveal the connection between ferroptosis and cellular senescence, Zhou et al. create a molecular regulatory network of "cellular senescence-ferroptosis escape" based on IFI16 (Zhou et al., 2025b). Whilst Chen et al. confirm elevated significantly Wnt5a expression in senescent tendon stem/progenitor cell, trigger the nonclassical Wnt signaling pathway. Functionally, Wnt5a inhibition

attenuated senescence, senescence-associated changes in cell polarity, and expression of SASP-related genes (Chen et al., 2021).

Conclusion

In summary, this study systematically analyzed the therapeutic effects of Feikang prescription in CS-induced COPD mice and preliminarily revealed its potential for treating COPD through multiple components, targets, and pathways. A variety of methods and *in vivo* experiments further validated that Feikang prescription alleviated airway inflammation-oxidative stress-ferroptosis and cell senescence. The findings indicated that Feikang prescription might regulated GPX4/Wnt5a/p53 signaling pathway to attenuate airway injury in COPD (Fig. 12). While GPX4-dependent ferroptosis modulated Wnt5a/p53-mediated cell senescence or Wnt5a/p53-mediated cell senescence regulated GPX4-dependent ferroptosis was still unclear. Moreover, the lack of *in vitro* experiments in cell hinders further elucidation of the underlying mechanism. Future research is necessary to determine the therapeutic potential mechanism by which Wnt5a/p53-mediated cell senescence may be regulated by GPX4-dependent ferroptosis, as well as the practical application of Feikang prescription in the treatment of COPD.

Data availability statements

All data were generated in-house, and no paper mill was utilized, and all data were produced internally. Each author pledges to take responsibility for every facet of the work, guaranteeing its correctness and integrity.

Ethics approval and consent to participate

The Laboratory Animal Ethical Committee of the First Affiliated Hospital of Guangzhou University of Chinese Medicine (Approval No. TCMF₁-2021,079). We applied in August 31, 2021.

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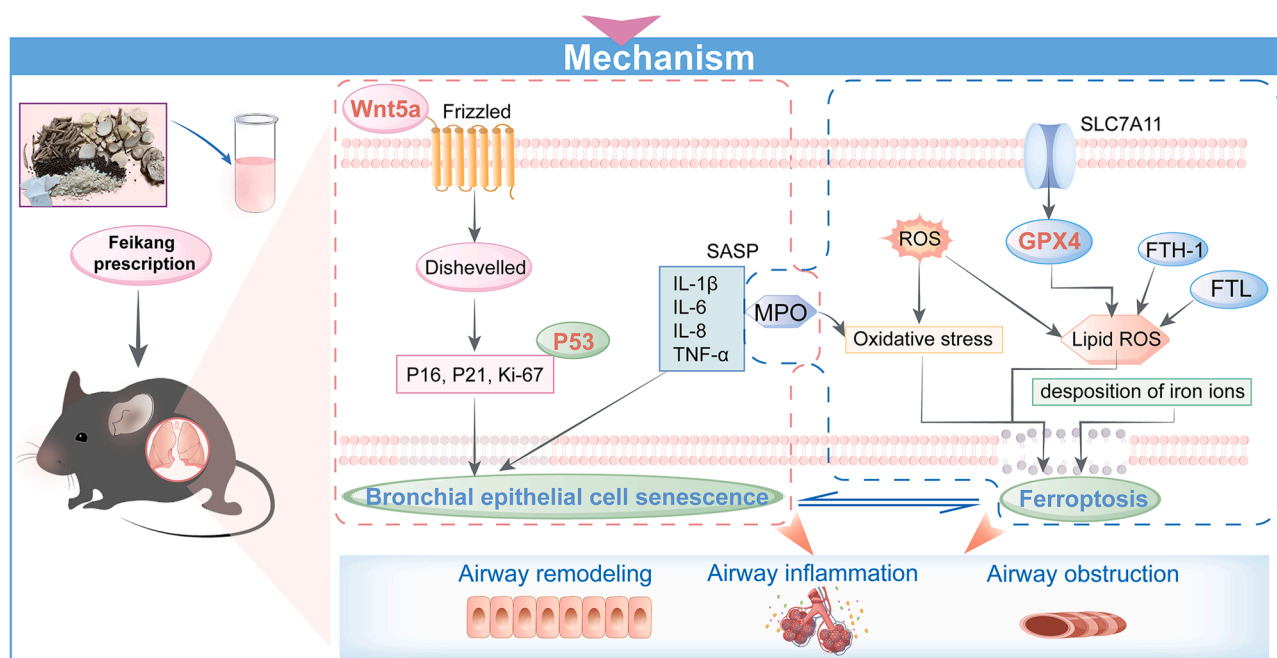


Fig. 12. The potential mechanism of Feikang prescription against COPD.

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CRediT authorship contribution statement

Xiaofang Li: Writing – review & editing, Writing – original draft, Visualization, Supervision, Software, Methodology, Formal analysis, Data curation. **Wenjiang Zheng:** Writing – review & editing, Visualization, Investigation, Formal analysis, Data curation. **Zhiyan Luo:** Visualization, Data curation. **Xinxin Zhang:** Visualization, Data curation. **Huiting Huang:** Visualization, Data curation. **Gang Liao:** Visualization, Data curation. **Huiqiu Liang:** Supervision. **Wujin Wen:** Supervision. **Yong Jiang:** Supervision, Resources. **Hongfa Zhuang:** Supervision, Resources. **Shaofeng Zhan:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare no financial or non-financial interests.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.phymed.2025.157175](https://doi.org/10.1016/j.phymed.2025.157175).

References

- Agustí, A., Celli, B.R., Criner, G.J., Halpin, D., Anzueto, A., Barnes, P., Bourbeau, J., Han, M.K., Martínez, F.J., Montes de Oca, M., Mortimer, K., Papi, A., Pavord, I., Roche, N., Salvi, S., Sin, D.D., Singh, D., Stockley, R., López Varela, M.V.,

- Wedzicha, J.A., Vogelmeier, C.F., 2023. Global initiative for chronic obstructive lung disease 2023 report: GOLD executive summary. *Eur. Respir. J.* 61. <https://doi.org/10.1183/13993003.00239-2023>.
- Bai, T., Guo, J., Deng, Y., Zheng, Y., Shang, J., Zheng, P., Liu, M., Yang, M., Zhang, J., 2024. A systematical strategy for quality markers screening of different methods processing platycodonis radix based on phytochemical analysis and the impact on chronic obstructive pulmonary disease. *J. Ethnopharmacol.* 319, 117311. <https://doi.org/10.1016/j.jep.2023.117311>.
- Bowin, C.F., Kozieliwicz, P., Grätz, L., Kowalski-Jahn, M., Schihada, H., Schulte, G., 2023. WNT stimulation induces dynamic conformational changes in the Frizzled-Dishevelled interaction. *Sci. Signal.* 16, eabo4974. <https://doi.org/10.1126/scisignal.abo4974>.
- Buekers, J., Delgado-Ortiz, L., Megaritis, D., Polhemus, A., Breuls, S., Buttery, S.C., Chynkiamis, N., Demeyer, H., Gimeno-Santos, E., Hume, E., Koch, S., Williams, P., Wuys, M., Hopkinson, N.S., Vogiatzis, I., Troosters, T., Frei, A., Garcia-Aymerich, J., 2024. Gait differences between COPD and healthy controls: systematic review and meta-analysis. *Eur. Respir. Rev.* 33. <https://doi.org/10.1183/16000617.0253-2023>.
- Chen, M., Li, Y., Xiao, L., Dai, G., Lu, P., Rui, Y., 2021. Noncanonical Wnt5a signaling regulates tendon stem/progenitor cells senescence. *Stem Cell. Res. Ther.* 12, 544. <https://doi.org/10.1186/s13287-021-02605-1>.
- Cheng, M., Yan, X., Wu, Y., Zeng, Z., Zhang, Y., Wen, F., Chen, J., Wang, T., 2025. Qingke Pingchuan granules alleviate airway inflammation in COPD exacerbation by inhibiting neutrophil extracellular traps in mice. *Phytomedicine* 136, 156283. <https://doi.org/10.1016/j.phymed.2024.156283>.
- Cui, S., Ye, J., 2024. 7-Dehydrocholesterol: a sterol shield against an iron sword. *Mol. Cell* 84, 1183–1185. <https://doi.org/10.1016/j.molcel.2024.03.003>.
- Deng, Y., Zhang, X., Li, R., Li, Z., Yang, B., Shi, P., Zhang, H., Wang, C., Wen, C., Li, G., Bian, L., 2022. Biomaterial-mediated presentation of wnt5a mimetic ligands enhances chondrogenesis and metabolism of stem cells by activating non-canonical Wnt signaling. *Biomaterials* 281, 121316. <https://doi.org/10.1016/j.biomaterials.2021.121316>.
- Devulder, J.V., 2024. Unveiling mechanisms of lung aging in COPD: a promising target for therapeutics development. *Chin. Med. J. Pulm. Crit. Care Med.* 2, 133–141. <https://doi.org/10.1016/j.pccm.2024.08.007>.
- Di Micco, R., Krizhanovsky, V., Baker, D., d'Adda di Fagagna, F., 2021. Cellular senescence in ageing: from mechanisms to therapeutic opportunities. *Nat. Rev. Mol. Cell Biol.* 22, 75–95. <https://doi.org/10.1038/s41580-020-00314-w>.
- Domingo, C., Busse, W.W., Hanania, N.A., Ertugrul, M., Millette, L.A., Maio-Twofoot, T., Jaumont, X., Palomares, O., 2025. The direct and indirect role of IgE on airway epithelium in asthma. *Allergy* 80, 919–931. <https://doi.org/10.1111/all.16459>.
- Dransfield, M., Kalhan, R., Stolz, D., 2023. Pushing (for) GOLD. *Eur. Respir. J.* 61. <https://doi.org/10.1183/13993003.00366-2023>.
- Fan, L., Bin, W., Ma, J., Ye, Z., Nie, X., Cheng, M., Xie, Y., Gu, P., Zhang, Y., You, X., Zhou, Y., Chen, W., 2023. Role and mechanism of WNT5A in benzo(a)pyrene-induced acute lung injury and lung function decline. *J. Hazard. Mater.* 460, 132391. <https://doi.org/10.1016/j.jhazmat.2023.132391>.
- Fan, Y., Wang, Y., Dan, W., Zhang, Y., Nie, L., Ma, Z., Zhuang, Y., Liu, B., Li, M., Liu, T., Wang, Z., Ye, L., Wei, Y., Lei, Y., Guo, C., An, J., Wang, C., Zhang, Y., Zeng, J., Wei, W., Gan, B., Li, L., 2025. PRMT5-mediated arginine methylation stabilizes GPX4 to suppress ferroptosis in cancer. *Nat. Cell Biol.* 27, 641–653. <https://doi.org/10.1038/s41556-025-01610-3>.

- Feng, L., Sun, J., Xia, L., Shi, Q., Hou, Y., Zhang, L., Li, M., Fan, C., Sun, B., 2024. Ferroptosis mechanism and Alzheimer's disease. *Neural Regen. Res.* 19, 1741–1750. <https://doi.org/10.4103/1673-5374.389362>.
- Freitas, F.P., Alborzinia, H., Dos Santos, A.F., Nepachalovich, P., Pedrera, L., Zilka, O., Inague, A., Klein, C., Aroua, N., Kaushal, K., Kast, B., Lorenz, S.M., Kunz, V., Nehring, H., Xavier da Silva, T.N., Chen, Z., Atici, S., Doll, S.G., Schaefer, E.L., Ekpo, I., Schmitz, W., Horling, A., Imming, P., Miyamoto, S., Wehman, A.M., Genaro-Mattos, T.C., Mirnics, K., Kumar, L., Klein-Seetharaman, J., Meierjohann, S., Weigand, I., Kroiss, M., Bornkamm, G.W., Gomes, F., Netto, L.E.S., Sathian, M.B., Konrad, D.B., Covey, D.F., Michalke, B., Bommert, K., Bargou, R.C., Garcia-Saez, A., Pratt, D.A., Fedorova, M., Trumpp, A., Conrad, M., Friedmann Angeli, J.P., 2024. 7-Dehydrocholesterol is an endogenous suppressor of ferroptosis. *Nature* 626, 401–410. <https://doi.org/10.1038/s41586-023-06878-9>.
- Fu, H.J., Zhou, X.Y., Qin, D.L., Qiao, Q., Wang, Q.Z., Li, S.Y., Zhu, Y.F., Li, Y.P., Zhou, J.M., Cai, H., Huang, F.H., Yu, L., Wang, L., Wu, A.G., Wu, J.M., Zhou, X.G., 2025. Inhibition of ferroptosis delays aging and extends healthspan across *Multiple species*. *Adv. Sci.* 12, e2416559. <https://doi.org/10.1002/adv.202416559> (Weinh).
- Gao, J., Shen, Y., Chen, Z., 2024. Novel insights into the pathological features of COPD: focus on oxidative stress and mitophagy. 4, e343. [10.1002/ctd2.343](https://doi.org/10.1002/ctd2.343).
- Giofrè, S.V., Napoli, E., Iraci, N., Speciale, A., Cimino, F., Muscarà, C., Molonia, M.S., Ruberto, G., Saija, A., 2021. Interaction of selected terpenoids with two SARS-CoV-2 key therapeutic targets: an in silico study through molecular docking and dynamics simulations. *Comput. Biol. Med.* 134, 104538. <https://doi.org/10.1016/j.combiomed.2021.104538>.
- GOLD., 2025. Global initiative for chronic obstructive lung disease (GOLD). Global Strategy for Prevention, Diagnosis and Management of COPD: 2025 Report. GOLD. <https://goldcopd.org/2024-gold-report/>.
- Gu, P., Wei, R., Liu, R., Yang, Q., He, Y., Guan, J., He, W., Li, J., Zhao, Y., Xie, L., He, J., Guo, Q., Hu, J., Bao, J., Wang, W., Guo, J., Zeng, Z., Chen, Z., Jiang, Y., Liu, Z., Chen, P., 2025. Aging-induced alternation in the gut microbiota impairs host antibacterial defense. *Adv. Sci.* 12, e2411008. <https://doi.org/10.1002/adv.202411008> (Weinh).
- Jiang, J., Zheng, Z., Chen, S., Liu, J., Jia, J., Huang, Y., Liu, Q., Cheung, C.Y., Sin, D.D., Yang, T., Wang, C., 2024. Hypoxia inducible factor (HIF) 3α prevents COPD by inhibiting alveolar epithelial cell ferroptosis via the HIF-3α-GPx4 axis. *Theranostics* 14, 5512–5527. <https://doi.org/10.1016/j.thno.99237>.
- Kaur, G., Muthumalage, T., Rahman, I., 2023. Clearance of senescent cells reverts the cigarette smoke-induced lung senescence and airspace enlargement in p16-3MR mice. *Aging Cell* 22, e13850. <https://doi.org/10.1111/ace1.13850>.
- Lee, J., Roh, J.L., 2024. Cholesterol-ferroptosis nexus: unveiling novel cancer therapeutic avenues. *Cancer Lett.* 597, 217046. <https://doi.org/10.1016/j.canlet.2024.217046>.
- Li, J., Xie, Y., Zhao, P., Qin, Y., Oliver, B.G., Tian, Y., Li, S., Wang, M., Liu, X., 2021. A Chinese herbal formula ameliorates COPD by inhibiting the inflammatory response via downregulation of p65, JNK, and p38. *Phytomedicine* 83, 153475. <https://doi.org/10.1016/j.phymed.2021.153475>.
- Li, L., Wang, X., 2022. Ferroptosis-associated cholesterol metabolism regulated by p85α in human bronchial epithelial cells with smoking. 2, e30. [10.1002/ctd2.30](https://doi.org/10.1002/ctd2.30).
- Li, Y., Ran, Q., Duan, Q., Jin, J., Wang, Y., Yu, L., Wang, C., Zhu, Z., Chen, X., Wang, L., Li, Z., Wang, J., Wu, Q., Wang, H., Tian, H., Song, S., Shan, Z., Zhai, Q., Qin, H., Chen, S., Fang, L., Yin, H., Zhou, H., Jiang, X., Wang, P., 2024. 7-Dehydrocholesterol dictates ferroptosis sensitivity. *Nature* 626, 411–418. <https://doi.org/10.1038/s41586-023-06983-9>.
- Liu, C., Liao, W., Chen, J., Yu, K., Wu, Y., Zhang, S., Chen, M., Chen, F., Wang, S., Cheng, T., Wang, J., Du, C., 2023. Cholesterol confers ferroptosis resistance onto myeloid-biased hematopoietic stem cells and prevents irradiation-induced myelosuppression. *Redox Biol.* 62, 102661. <https://doi.org/10.1016/j.redox.2023.102661>.
- Liu, J., Sun, T., Liu, S., Liu, J., Fang, S., Tan, S., Zeng, Y., Zhang, B., Li, W., 2022a. Dissecting the molecular mechanism of cepharanthine against COVID-19, based on a network pharmacology strategy combined with RNA-sequencing analysis, molecular docking, and molecular dynamics simulation. *Comput. Biol. Med.* 151, 106298. <https://doi.org/10.1016/j.combiomed.2022.106298>.
- Liu, J., Xiao, Q., Xiao, J., Niu, C., Li, Y., Zhang, X., Zhou, Z., Shu, G., Yin, G., 2022b. Wnt/β-catenin signalling: function, biological mechanisms, and therapeutic opportunities. *Signal. Transduct. Target. Ther.* 7, 3. <https://doi.org/10.1038/s41392-021-00762-6>.
- Liu, Y., Gu, W., 2022. p53 in Ferroptosis regulation: the new weapon for the old guardian. *Cell Death. Differ.* 29, 895–910. <https://doi.org/10.1038/s41418-022-00943-y>.
- Lu, Y., Zhu, F., Zhou, X., Li, Y., Rong, G., Liu, N., Hong, J., Cheng, Y., 2025. A supramolecular deferoxamine-crisaborole nanoparticle targets ferroptosis, inflammation, and oxidative stress in the treatment of retinal ischemia/reperfusion injury. *Nano Lett.* 25, 1058–1066. <https://doi.org/10.1021/acs.nanolett.4c05012>.
- Mastelaro de Rezende, M., Zenker Justo, G., Julian Paredes-Gamero, E., Gosens, R., 2020. Wnt-5A/B signaling in hematopoiesis throughout life. *Cells* 9. <https://doi.org/10.3390/cells9081801>.
- Miller, K.N., Li, B., Pierce-Hoffman, H.R., Patel, S., Lei, X., Rajesh, A., Teneche, M.G., Havas, A.P., Gandhi, A., Macip, C.C., Lyu, J., Victorelli, S.G., Woo, S.H., Lagnado, A. B., LaPorta, M.A., Liu, T., Dasgupta, N., Li, S., Davis, A., Korotkov, A., Hultenius, E., Gao, Z., Altman, Y., Porritt, R.A., Garcia, G., Mogler, C., Seluanov, A., Gorbunova, V., Kaech, S.M., Tian, X., Dou, Z., Chen, C., Passos, J.F., Adams, P.D., 2025. p53 Enhances DNA repair and suppresses cytoplasmic chromatin fragments and inflammation in senescent cells. *Nat. Commun.* 16, 2229. <https://doi.org/10.1038/s41467-025-57229-3>.
- Monk, B.A., Mill, C.E.J., Angelini, G.D., Johnson, J.L., George, S.J., 2015. Wnt5a-mediated rescue of vascular smooth muscle cells (VSMCs) from oxidative stress induced apoptosis is retarded with ageing. *Atherosclerosis* 241, e77–e78. <https://doi.org/10.1016/j.atherosclerosis.2015.04.271>.
- Pang, L., Zhao, Y., Xu, Y., Gao, C., Wang, C., Yu, X., Wang, F., He, K., 2024. Mechanisms underlying the therapeutic effects of JianPiYiFei II granules in treating COPD based on GEO datasets, network pharmacology, molecular docking, and molecular dynamics simulations. *Biology* 13. <https://doi.org/10.3390/biology13090711> (Basel).
- Paudel, K.R., Panth, N., Manandhar, B., Singh, S.K., Gupta, G., Wich, P.R., Nammi, S., MacLoughlin, R., Adams, J., Warkiani, M.E., Chellappan, D.K., Oliver, B.G., Hansbro, P.M., Dua, K., 2022. Attenuation of cigarette-smoke-induced oxidative stress, senescence, and inflammation by berberine-loaded liquid crystalline nanoparticles: *in vitro* study in 16HBE and RAW264.7 cells. *Antioxidants* 11. <https://doi.org/10.3390/antiox11050873> (Basel).
- Peng, X., Sun, B., Tang, C., Shi, C., Xie, X., Wang, X., Jiang, D., Li, S., Jia, Y., Wang, Y., Tang, H., Zhong, S., Piao, M., Cui, X., Zhang, S., Wang, F., Wang, Y., Na, R., Huang, R., Jiang, Y., Zhang, W., Xu, J., Lin, K., Guo, J., Pan, Z., Wang, K., Zhao, Q., Liu, H., Yu, B., Ji, Y., Zhang, J., Li, S., Tian, J., 2025. HMOX1-LDHB interaction promotes ferroptosis by inducing mitochondrial dysfunction in foamy macrophages during advanced atherosclerosis. *Dev. Cell.* 60, 1070–1086.e1078. <https://doi.org/10.1016/j.devcel.2024.12.011>.
- Rodencal, J., Kim, N., He, A., Li, V.L., Lange, M., He, J., Tarangelo, A., Schaefer, Z.T., Olzmann, J.H., Long, J.Z., Sage, J., Dixon, S.J., 2024. Sensitization of cancer cells to ferroptosis coincident with cell cycle arrest. *Cell Chem. Biol.* 31, 234–248.e213. <https://doi.org/10.1016/j.chembiol.2023.10.011>.
- Shao, C., Yuan, J., Liu, Y., Qin, Y., Wang, X., Gu, J., Chen, G., Zhang, B., Liu, H.K., Zhao, J., Zhu, H.L., Qian, Y., 2020. Epileptic brain fluorescent imaging reveals apigenin can relieve the myeloperoxidase-mediated oxidative stress and inhibit ferroptosis. *Proc. Natl. Acad. Sci. U. S. A.* 117, 10155–10164. <https://doi.org/10.1073/pnas.1917946117>.
- Sun, D.Y., Wu, W.B., Wu, J.J., Shi, Y., Xu, J.J., Ouyang, S.X., Chi, C., Shi, Y., Ji, Q.X., Miao, J.H., Fu, J.T., Tong, J., Zhang, P.P., Zhang, J.B., Li, Z.Y., Qu, L.F., Shen, F.M., Li, D.J., Wang, P., 2024. Pro-ferroptotic signaling promotes arterial aging via vascular smooth muscle cell senescence. *Nat. Commun.* 15, 1429. <https://doi.org/10.1038/s41467-024-45823-w>.
- Sun, Q., Liu, D., Cui, W., Cheng, H., Huang, L., Zhang, R., Gu, J., Liu, S., Zhuang, X., Lu, Y., Chu, B., Li, J., 2023. Cholesterol mediated ferroptosis suppression reveals essential roles of Coenzyme Q and squalene. *Commun. Biol.* 6, 1108. <https://doi.org/10.1038/s42003-023-05477-8>.
- Sun, Y., Xue, Z., Huang, T., Che, X., Wu, G., 2022. Lipid metabolism in ferroptosis and ferroptosis-based cancer therapy. *Front. Oncol.* 12, 941618. <https://doi.org/10.3389/fonc.2022.941618>.
- Sun, Z., Li, L., Zhang, L., 2025. Apigenin enhancing oxidative resistance and proteostasis to extend lifespan via PTEN-mediated AKT signalling pathway. *Biochim. Biophys. Acta Mol. Basis. Dis.* 1871, 167670. <https://doi.org/10.1016/j.bbdis.2025.167670>.
- Suryadevara, V., Hudgins, A.D., Rajesh, A., Pappalardo, A., Karpova, A., Dey, A.K., Hertz, A., Agudelo, A., Rocha, A., Soygar, B., Schilling, B., Carver, C.M., Aguayo-Mazzucato, C., Baker, D.J., Bernlohr, D.A., Jurk, D., Mangarova, D.B., Quardokus, E. M., Enninga, E.A.L., Schmidt, E.L., Chen, F., Duncan, F.E., Cambuli, F., Kaur, G., Kuchel, G.A., Lee, G., Daldrop-Link, H.E., Martini, H., Phatnani, H., Al-Naggar, I.M., Rahman, I., Nie, J., Passos, J.F., Silverstein, J.C., Campisi, J., Wang, J., Iwasaki, K., Barbosa, K., Metis, K., Nerke, K., Niederhofer, L.J., Ding, L., Wang, L., Adams, L. C., Ruiyang, L., Doolittle, M.L., Teneche, M.G., Schafer, M.J., Xu, M., Hajipour, M., Boroumand, M., Basisty, N., Sloan, N., Slavov, N., Kuksenko, O., Robson, P., Gomez, P.T., Vasilikos, P., Adams, P.D., Carapeto, P., Zhu, Q., Ramasamy, R., Perez-Lorenzo, R., Fan, R., Dong, R., Montgomery, R.R., Shaikh, S., Vickovic, S., Yin, S., Kang, S., Suvakov, S., Khosla, S., Garovic, V.D., Menon, V., Xu, Y., Song, Y., Suh, Y., Dou, Z., Neretti, N., 2024. SenNet recommendations for detecting senescent cells in different tissues. *Nat. Rev. Mol. Cell Biol.* 25, 1001–1023. <https://doi.org/10.1038/s41580-024-00738-8>.
- Tang, D., Chen, X., Kang, R., Kroemer, G., 2021. Ferroptosis: molecular mechanisms and health implications. *Cell Res.* 31, 107–125. <https://doi.org/10.1038/s41422-020-00441-1>.
- Wang, B., Han, J., Elisseeff, J.H., Demaria, M., 2024. The senescence-associated secretory phenotype and its physiological and pathological implications. *Nat. Rev. Mol. Cell Biol.* 25, 958–978. <https://doi.org/10.1038/s41580-024-00727-x>.
- Wang, C., Zhou, J., Wang, J., Li, S., Fukunaga, A., Yodoi, J., Tian, H., 2020. Progress in the mechanism and targeted drug therapy for COPD. *Signal. Transduct. Target. Ther.* 5, 248. <https://doi.org/10.1038/s41392-020-00345-x>.
- Wang, J., Yue, H., Dong, Y., Liu, T., Li, J., 2025a. Effective compound combination of Bufe Yishen formula ameliorates PM2.5-induced COPD by inhibiting mitochondrial oxidative stress through SIRT3-mediated FOXO3 deacetylation. *Phytomedicine* 140, 156568. <https://doi.org/10.1016/j.phymed.2025.156568>.
- Wang, X., Murugesan, P., Zhang, P., Xu, S., Peng, L., Wang, C., Cai, H., 2022. NADPH oxidase isoforms in COPD patients and acute cigarette smoke-exposed mice: induction of oxidative stress and lung inflammation. *Antioxidants* 11. <https://doi.org/10.3390/antiox11081539>. (Basel).
- Wang, X., Wu, Q., Zhong, M., Chen, Y., Wang, Y., Li, X., Zhao, W., Ge, C., Wang, X., Yu, Y., Yang, S., Wang, T., Xie, E., Shi, W., Yin, J., Wang, F., 2025b. Adipocyte-derived ferroptotic signaling mitigates obesity. *Cell Metab.* 37, 673–691.e677. <https://doi.org/10.1016/j.cmet.2024.11.010>.
- Wechsler, M.E., Wells, J.M., 2024. What every clinician should know about inflammation in COPD. *ERJ Open. Res.* 10. <https://doi.org/10.1183/23120541.00177-2024>.
- Xiaofei, Y., Tingting, L., Xuan, W., Zhiyi, H., 2022. Erythromycin attenuates oxidative stress-induced cellular senescence via the PI3K-mTOR signaling pathway in chronic obstructive pulmonary disease. *Front. Pharmacol.* 13, 1043474. <https://doi.org/10.3389/fphar.2022.1043474>.

- Xie, D., Quan, J., Yu, X., Liang, Z., Chen, Y., Wu, L., Lin, L., Fan, L., 2024. Molecular mechanism of Jianpiyifei II granules in the treatment of chronic obstructive pulmonary disease: network pharmacology analysis, molecular docking, and experimental assessment. *Phytomedicine* 126, 155273. <https://doi.org/10.1016/j.phymed.2023.155273>.
- Xie, J., Liu, M., Gao, Y., Liu, C., Wu, F., Tong, J., Li, Z., Zhu, J., 2023. Integration of metabolomics and network pharmacology to reveal the protective mechanism underlying Qibai Pingfei capsule on chronic obstructive pulmonary disease. *Front. Pharmacol.* 14, 1258138. <https://doi.org/10.3389/fphar.2023.1258138>.
- Xu, Y., Zhang, L., Chen, C., Zou, M., Wang, K., Liu, X., Kang, T., Li, M., Wu, D., Jiang, Z., Liu, J., 2025. Investigation of the efficacy and potential pharmacological mechanism of Yupingfeng in treating chronic obstructive pulmonary disease: a meta-analysis and in silico study. *J. Ethnopharmacol.* 343, 119441. <https://doi.org/10.1016/j.jep.2025.119441>.
- Yan, H.F., Zou, T., Tuo, Q.Z., Xu, S., Li, H., Belaidi, A.A., Lei, P., 2021. Ferroptosis: mechanisms and links with diseases. *Signal. Transduct. Target. Ther.* 6, 49. <https://doi.org/10.1038/s41392-020-00428-9>.
- Yang, L., Wen, M., Liu, X., Wang, K., Wang, Y., 2020. Feikang granules ameliorate pulmonary inflammation in the rat model of chronic obstructive pulmonary disease via TLR2/4-mediated NF- κ B pathway. *BMC Complement. Med. Ther.* 20, 170. <https://doi.org/10.1186/s12906-020-02964-x>.
- Yang, Y., Shen, W., Zhang, Z., Dai, Y., Zhang, Z., Liu, T., Yu, J., Huang, S., Ding, Y., You, R., Wang, Z., Wu, Y., Bian, T., 2025. FSP1 acts in parallel with GPX4 to inhibit ferroptosis in chronic obstructive pulmonary disease. *Am. J. Respir. Cell Mol. Biol.* 72, 551–562. <https://doi.org/10.1165/rcmb.2023-0467OC>.
- Yi, C., Sun, X., Lin, Y., Gu, C., Ding, L., Lu, X., Yang, Z., Zhang, Y., Ma, L., Gu, W., Qu, A., Zhou, X., Li, X., Xu, J., Ling, Z., Xie, Y., Lu, H., Sun, B., 2021. Comprehensive mapping of binding hot spots of SARS-CoV-2 RBD-specific neutralizing antibodies for tracking immune escape variants. *Genome Med.* 13, 164. <https://doi.org/10.1186/s13073-021-00985-w>.
- Yoshida, M., Minagawa, S., Araya, J., Sakamoto, T., Hara, H., Tsubouchi, K., Hosaka, Y., Ichikawa, A., Saito, N., Kadota, T., Sato, N., Kurita, Y., Kobayashi, K., Ito, S., Utsumi, H., Wakui, H., Numata, T., Kaneko, Y., Mori, S., Asano, H., Yamashita, M., Odaka, M., Morikawa, T., Nakayama, K., Iwamoto, T., Imai, H., Kuwano, K., 2019. Involvement of cigarette smoke-induced epithelial cell ferroptosis in COPD pathogenesis. *Nat. Commun.* 10, 3145. <https://doi.org/10.1038/s41467-019-10991-7>.
- Zeng, X.X., Gao, D.D., Zhang, F., 2024. Cortex Mori Radicis[Morus Alba L. (Moraceae)] extract alleviates senescence via PI3K/Akt signaling in COPD fibroblasts. *Phytomedicine* 135, 156004. <https://doi.org/10.1016/j.phymed.2024.156004>.
- Zhan, Y., Huang, Q., Deng, Z., Chen, S., Yang, R., Zhang, J., Zhang, Y., Peng, M., Wu, J., Gu, Y., Zeng, Z., Xie, J., 2025. DNA hypomethylation-mediated upregulation of GADD45B facilitates airway inflammation and epithelial cell senescence in COPD. *J. Adv. Res.* 68, 201–214. <https://doi.org/10.1016/j.jare.2024.02.005>.
- Zhang, J., Wei, Y., Yue, Y., Jiao, H., Wu, Y., Fu, W., Lin, K.M., Lu, C., Mou, S., Zhong, Q., 2024a. RIPK4 promotes oxidative stress and ferroptotic death through the downregulation of ACSM1. *Proc. Natl. Acad. Sci. U. S. A.* 121, e2410628121. <https://doi.org/10.1073/pnas.2410628121>.
- Zhang, Q., Liu, J., Duan, H., Li, R., Peng, W., Wu, C., 2021. Activation of Nrf2/HO-1 signaling: an important molecular mechanism of herbal medicine in the treatment of atherosclerosis via the protection of vascular endothelial cells from oxidative stress. *J. Adv. Res.* 34, 43–63. <https://doi.org/10.1016/j.jare.2021.06.023>.
- Zhang, W., Liu, Y., Liao, Y., Zhu, C., Zou, Z., 2024b. GPX4, ferroptosis, and diseases. *Biomed. Pharmacother.* 174, 116512. <https://doi.org/10.1016/j.biopha.2024.116512>.
- Zhou, C., Qin, Y., Zhao, W., Liang, Z., Li, M., Liu, D., Bai, L., Chen, Y., Chen, Y., Cheng, Y., Chu, T., Chu, Q., Deng, H., Dong, Y., Fang, W., Fu, X., Gao, B., Han, Y., He, Y., Hong, Q., Hu, J., Hu, Y., Jiang, L., Jin, Y., Lan, F., Li, Q., Li, S., Li, W., Li, Y., Liang, W., Lin, G., Lin, X., Liu, M., Liu, X., Liu, X., Liu, Z., Lv, T., Mu, C., Ouyang, M., Qin, J., Ren, S., Shi, H., Shi, M., Su, C., Su, J., Sun, D., Sun, Y., Tang, H., Wang, H., Wang, K., Wang, K., Wang, M., Wang, Q., Wang, W., Wang, X., Wang, Y., Wang, Z., Wang, Z., Wu, L., Wu, D., Xie, B., Xie, M., Xie, X., Xie, Z., Xu, S., Xu, X., Yang, X., Yin, Y., Yu, Z., Zhang, J., Zhang, J., Zhang, J., Zhang, X., Zhang, Y., Zhong, D., Zhou, Q., Zhou, X., Zhou, Y., Zhu, B., Zhu, Z., Zou, C., Zhong, N., He, J., Bai, C., Hu, C., Li, W., Song, Y., Zhou, J., Han, B., Varga, J., Barreiro, E., Park, H.Y., Petrella, F., Saito, Y., Goto, T., Igai, H., Bravaccini, S., Zanon, M., Solli, P., Watanabe, S., Fiorelli, A., Nakada, T., Ichiki, Y., Berardi, R., Tsoukalas, N., Girard, N., Rossi, A., Passaro, A., Hida, T., Li, S., Chen, L., Chen, R., 2023. International expert consensus on diagnosis and treatment of lung cancer complicated by chronic obstructive pulmonary disease. *Transl. Lung Cancer Res.* 12, 1661–1701. <https://doi.org/10.21037/tlcr-23-339>.
- Zhou, S., Liao, F., Wen, H., 2025a. Isoquercetin alleviates osteoarthritis via regulating the NOX4/Nrf2 redox imbalance in senescent chondrocytes. *Int. J. Biol. Macromol.* 306, 141562. <https://doi.org/10.1016/j.ijbiomac.2025.141562>.
- Zhou, Y., Zeng, L., Cai, L., Zheng, W., Liu, X., Xiao, Y., Jin, X., Bai, Y., Lai, M., Li, H., Jiang, H., Hu, S., Pan, Y., Zhang, J., Shao, C., 2025b. Cellular senescence-associated gene IFI16 promotes HMOX1-dependent evasion of ferroptosis and radioresistance in glioblastoma. *Nat. Commun.* 16, 1212. <https://doi.org/10.1038/s41467-025-56456-y>.
- Zhou, Z., Yao, J., Wu, D., Huang, X., Wang, Y., Li, X., Lu, Q., Qiu, Y., 2024. Type 2 cytokine signaling in macrophages protects from cellular senescence and organismal aging. *Immunity* 57, 513–527.e516. <https://doi.org/10.1016/j.immuni.2024.01.001>.
- Zhu, J., Li, X., Zhou, Y., Ge, C., Li, X., Hou, M., Wei, Y., Chen, Y., Leong, K.W., Yin, L., 2024. Inhaled immunoantimicrobials for the treatment of chronic obstructive pulmonary disease. *Sci. Adv.* 10, eabd7904. <https://doi.org/10.1126/sciadv.abd7904>.
- Zhu, L., Chen, D., Zhu, Y., Pan, T., Xia, D., Cai, T., Lin, H., Lin, J., Jin, X., Wu, F., Yu, S., Zhu, K., Xu, L., Chen, Y., 2021. GPX4-regulated ferroptosis mediates S100-induced experimental autoimmune hepatitis associated with the Nrf2/HO-1 signaling pathway. *Oxid. Med. Cell. Longev.* 2021, 6551069. <https://doi.org/10.1155/2021/6551069>.